

# CARTILAGE AND BONE

Histology

# OBJECTIVES

- Identify main functions of bone tissue
- Identify the major parts of a long bones
- Describe the cells found in bone tissue
- Describe the components of an osteon
- Compare and contrast intramembranous and endochondral ossification.
- Describe the growth process at the epiphyseal plate
- Describe appositional bone growth
- Describe bone remodeling in the context of a pathologic bone fracture

# Cartilage

- It is a **tough ,resilient** type of connective tissue that structurally supports certain soft tissues, notably in the respiratory tract, and provides cushioned, low-friction surfaces in joints.
- Cells of cartilage, **chondrocytes** , make up a small percentage of the tissue's mass, which is mainly a flexible mass of **extracellular matrix(ECM)** .
- Chondrocytes are embedded within **lacunae** surrounded by the ECM.
- Cartilage ECM typically includes **collagen** as well as abundant **proteoglycans** , notably **aggrecan** , which bind a large amount of **water** .
- Cartilage always **lacks blood vessels** , lymphatics, and nerves, but it is usually surrounded by a dense connective tissue **perichondrium** that is vascularized.
- There are three major forms of cartilage:
  - ▣ (1)**hyaline cartilage** ,
  - ▣ (2) **elastic cartilage** , and
  - ▣ (3) **fibrocartilage**

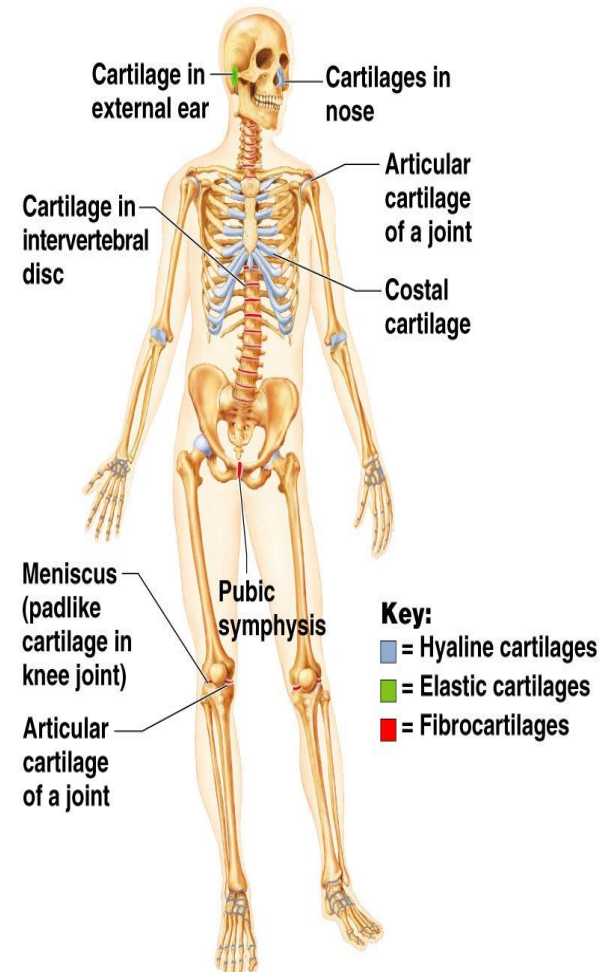
# SKELETAL CARTILAGE

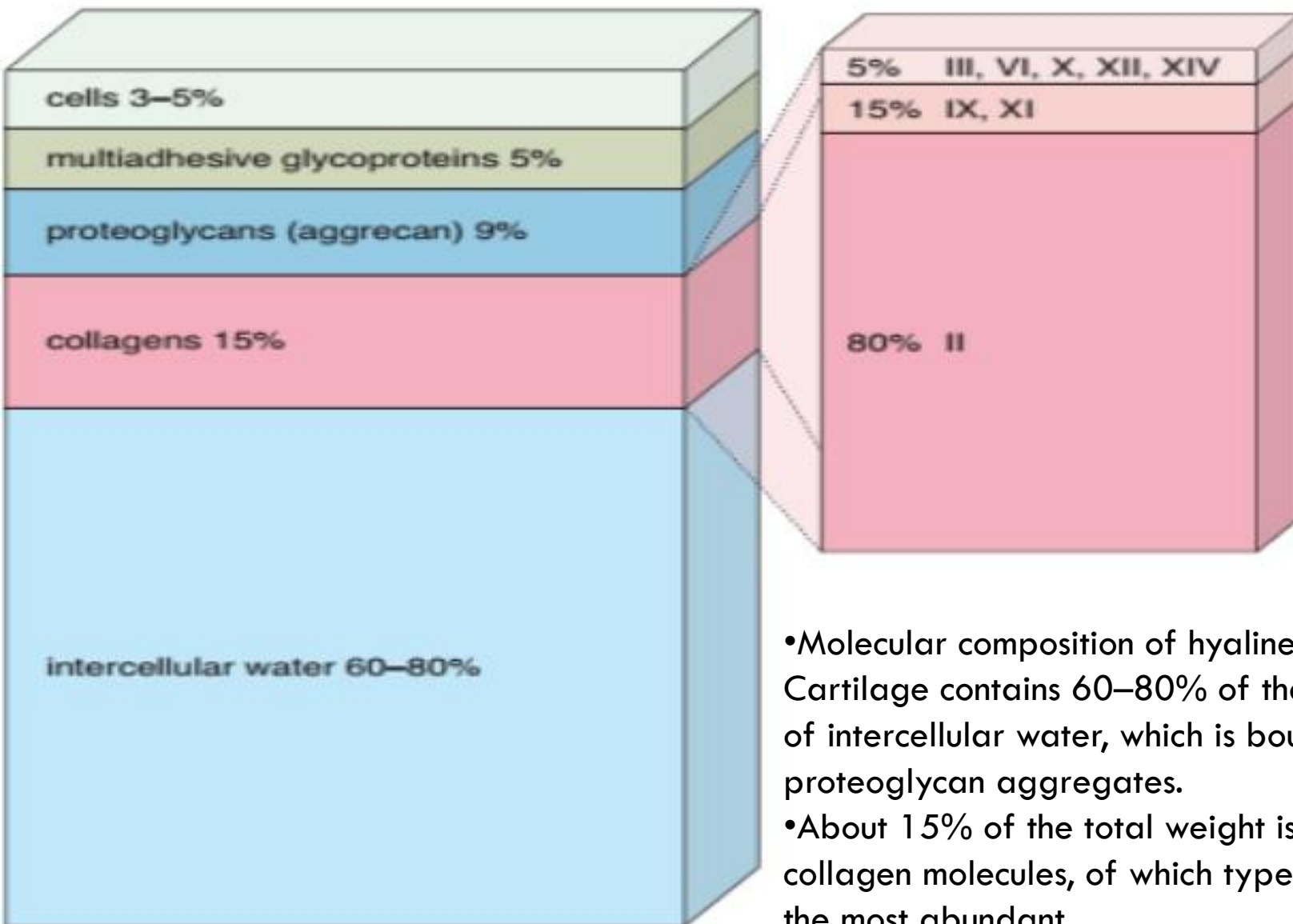
- **Chondrocytes:** cartilage producing cells.
- **Lacunae:** small cavities where the chondrocytes are encased.
- **Extracellular matrix:** jellylike ground substance.
- **Perichondrium:** layer of dense irregular connective tissue that surrounds the cartilage.
- No blood vessels or nerves



# TYPES OF CARTILAGE

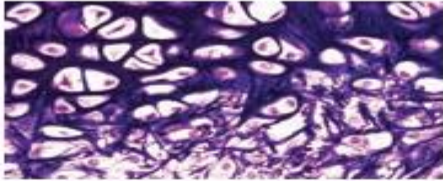
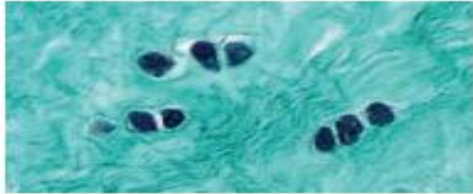
- Hyaline cartilage – (glassy)
  - ▣ Most abundant cartilage
  - ▣ Provides support through flexibility
  - ▣ Articular cartilages and costal cartilage, larynx, trachea, and nose
- Elastic cartilage – contains many elastic fibers
  - ▣ Able to tolerate repeated bending
  - ▣ Ear and epiglottis
- Fibrocartilage – resists strong compression and strong tension
  - ▣ An intermediate between hyaline and elastic cartilage
  - ▣ Intervertebral discs and pubic symphysis

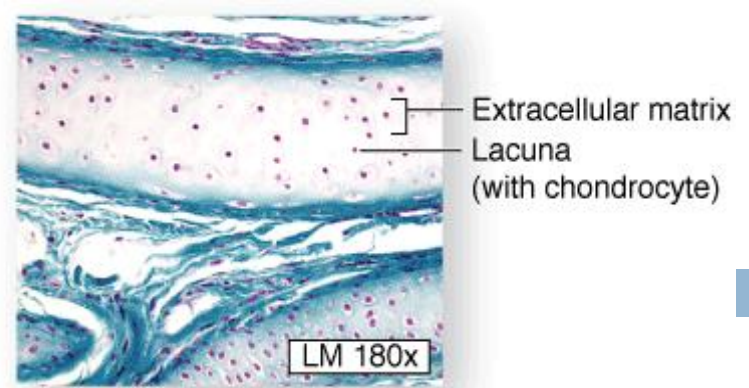
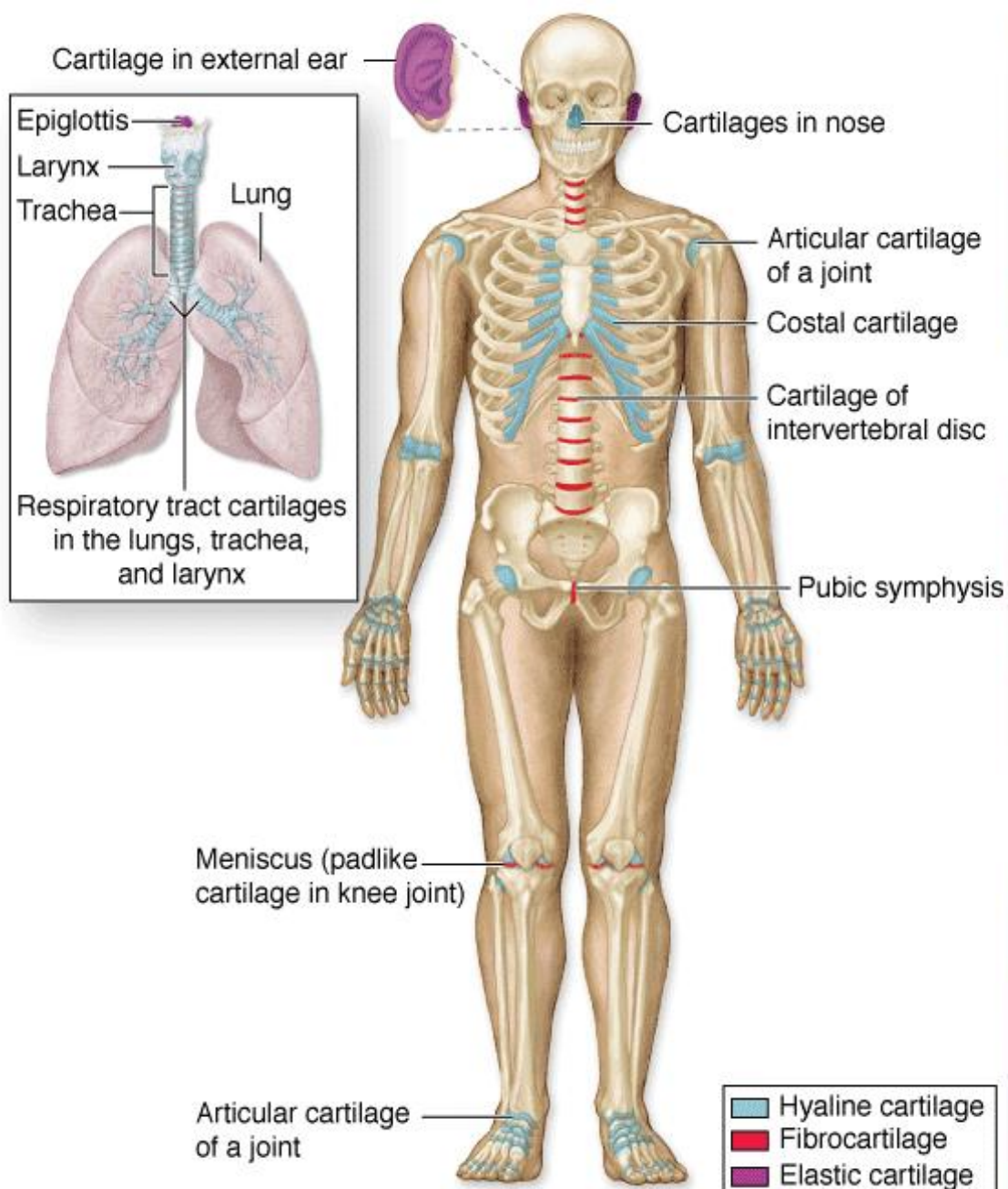




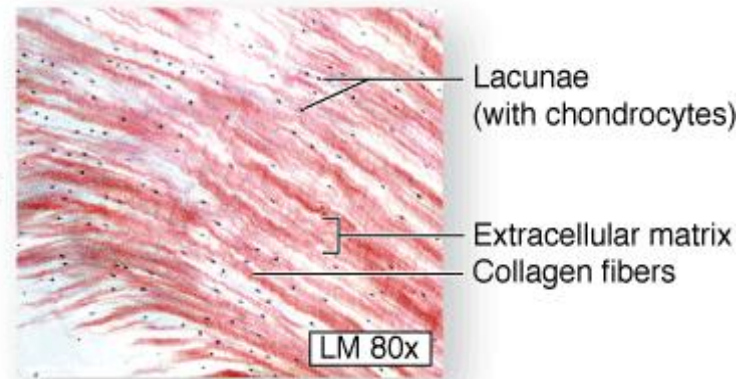
- Molecular composition of hyaline cartilage. Cartilage contains 60–80% of the wet weight of intercellular water, which is bound by proteoglycan aggregates.
- About 15% of the total weight is attributed to collagen molecules, of which type II collagen is the most abundant.
- Chondrocytes occupy only 3–5% of the total cartilage mass.

**TABLE 7.1 Summary of Cartilage Features**

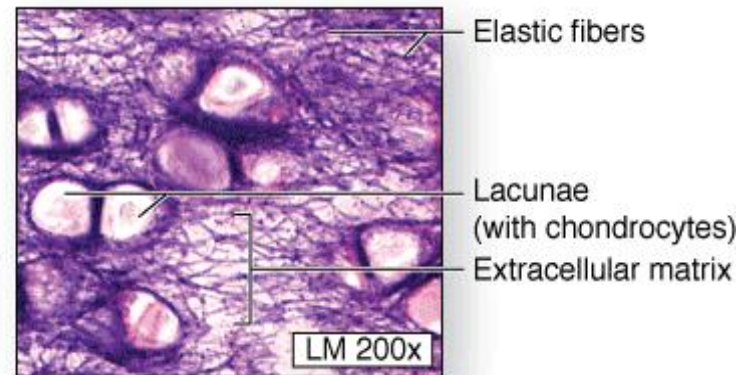
| Features                                        | Hyaline Cartilage                                                                                                                                                                                                                                                                         | Elastic Cartilage                                                                                                                                     | Fibrocartilage                                                                                                                                                                                      |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 |                                                                                                                                                                                                          |                                                                     |                                                                                                                  |
| Location                                        | Fetal skeletal tissue, epiphyseal plates, articular surface of synovial joints, costal cartilages of rib cage, cartilages of nasal cavity, larynx (thyroid, cricoid, and arytenoids), rings of trachea and plates in bronchi                                                              | Pinna of external ear, external acoustic meatus, auditory (Eustachian) tube, cartilages of larynx (epiglottis, corniculate, and cuneiform cartilages) | Intervertebral discs, symphysis pubis, articular discs (sternoclavicular and temporomandibular joints), menisci (knee joint), triangular fibrocartilage complex (wrist joint), insertion of tendons |
| Function                                        | Resists compression<br>Provides cushioning, smooth, and low-friction surface for joints<br>Provides structural support in respiratory system (larynx, trachea, and bronchi)<br>Forms foundation for development of fetal skeleton and further endochondral bone formation and bone growth | Provides flexible support                                                                                                                             | Resists deformation under stress                                                                                                                                                                    |
| Presence of perichondrium                       | Yes (except articular cartilage and epiphyseal plates)                                                                                                                                                                                                                                    | Yes                                                                                                                                                   | No                                                                                                                                                                                                  |
| Undergoes calcification                         | Yes (i.e., during endochondral bone formation, during aging process)                                                                                                                                                                                                                      | No                                                                                                                                                    | Yes (i.e., calcification of fibrocartilaginous callus during bone repair)                                                                                                                           |
| Main cell types present                         | Chondroblasts and chondrocytes                                                                                                                                                                                                                                                            | Chondroblasts and chondrocytes                                                                                                                        | Chondrocytes and fibroblasts                                                                                                                                                                        |
| Characteristic features of extracellular matrix | Type II collagen fibrils and aggrecan (the most important proteoglycan)                                                                                                                                                                                                                   | Type II collagen fibrils, elastic fibers, and aggrecan                                                                                                | Types I and II collagen fibers and versican (a proteoglycan secreted by fibroblasts)                                                                                                                |
| Growth                                          | Interstitially and appositionally, very limited in adults                                                                                                                                                                                                                                 |                                                                                                                                                       |                                                                                                                                                                                                     |



**b Hyaline cartilage**



**c Fibrocartilage**



**d Elastic cartilage**

# BONE FUNCTION

- Support
- Protection (protect internal organs)
- Movement (provide leverage system for skeletal muscles, tendons, ligaments and joints)
- Mineral homeostasis (bones act as reserves of minerals important for the body like calcium or phosphorus)
- Hematopoiesis: blood cell formation
- Storage of adipose tissue: yellow marrow

# Bone

- It is a type of connective tissue with a **calcified** extracellular matrix (ECM).
- It is specialized to:
  - ▣ **Support** the body,
  - ▣ **Protect** many internal organs, and
  - ▣ **Act** as the body's **Ca<sup>2</sup> + reservoir**

# SHAPE OF BONES

- Long bones (e.g., humerus, femur)
- Short bones (e.g., carpals, tarsals, patella)
- Flat bones (e.g., parietal bone, scapula, sternum)
- Irregular bones (e.g., vertebrae, hip bones)



# BONE ANATOMY

**Diaphysis:** long shaft of bone

**Epiphysis:** ends of bone

**Epiphyseal plate:** growth plate

**Metaphysis:** b/w epiphysis and diaphysis

**Articular cartilage:** covers epiphysis

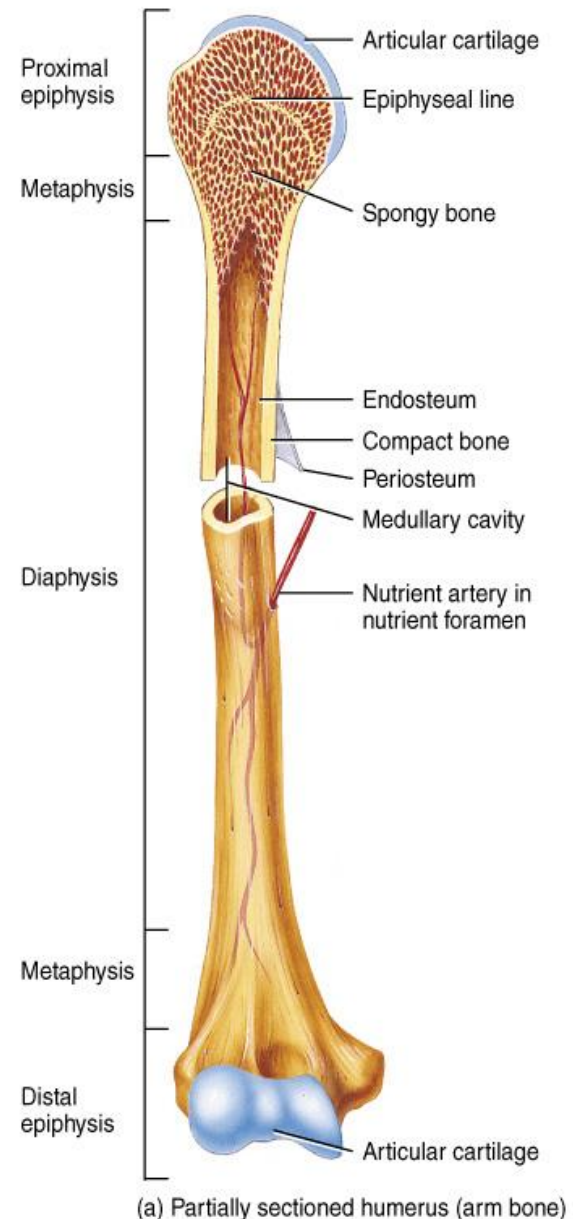
**Periosteum:** bone covering (pain sensitive)

**Sharpey's fibers:** periosteum attaches to underlying bone

**Medullary cavity:** Hollow chamber in bone

- red marrow produces blood cells
- yellow marrow is adipose

**Endosteum:** thin layer lining the medullary cavity

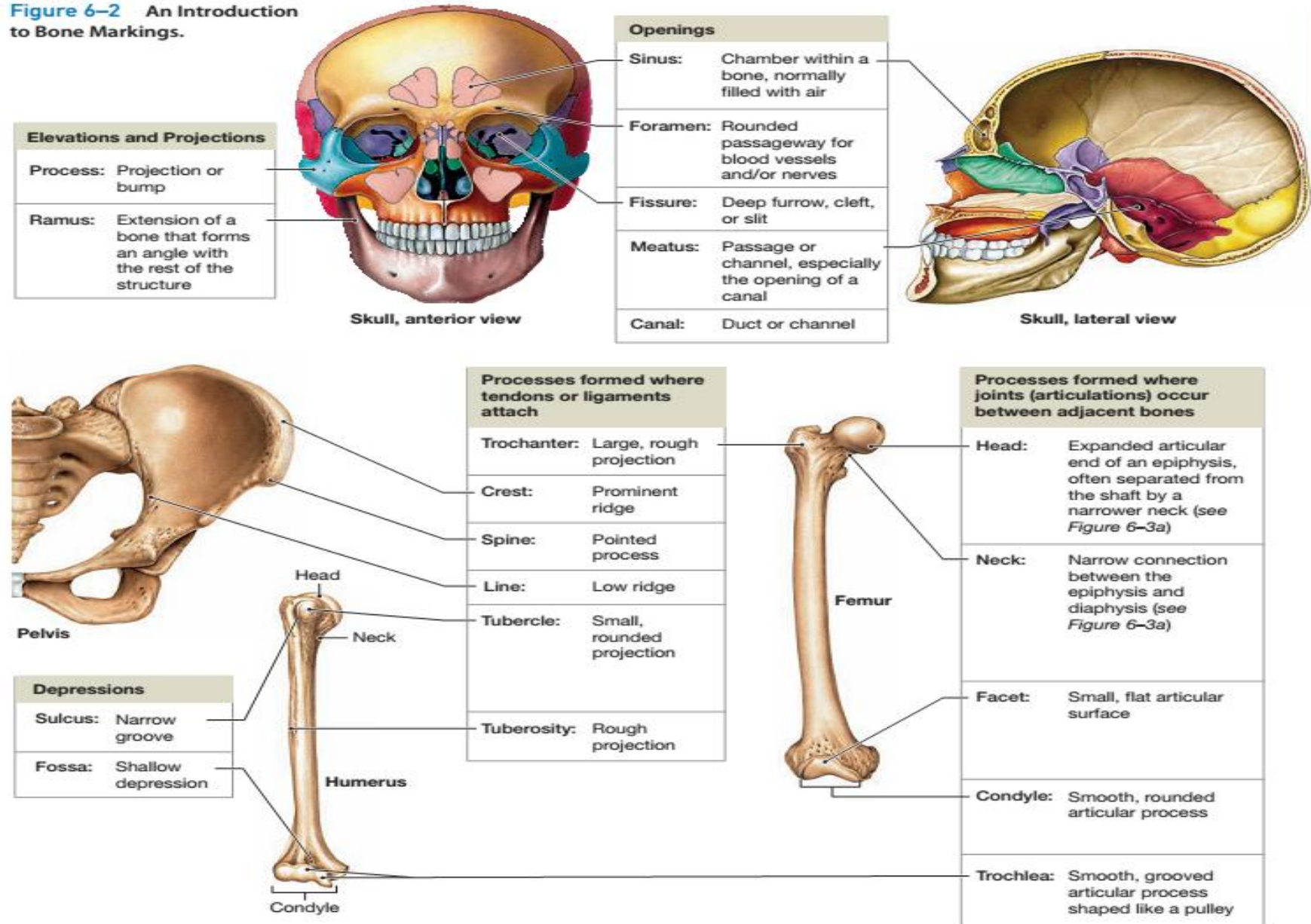


# Bone Markings

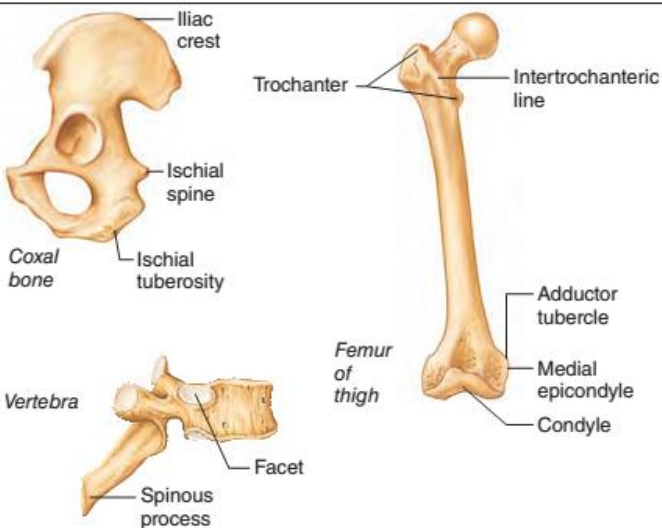
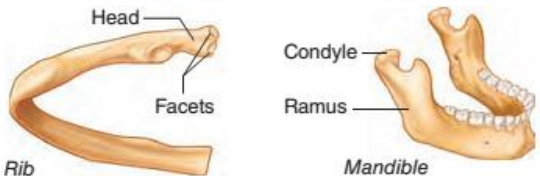
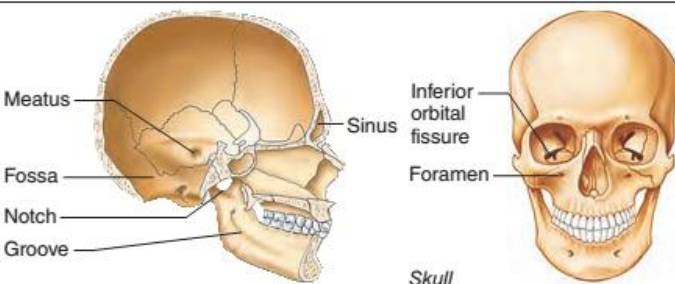


- The surfaces of each bone in your body have characteristic features.
- Elevations or projections form where tendons and ligaments attach, and where adjacent bones articulate (that is, at joints).
- Depressions, grooves, and tunnels in bone are sites where blood vessels or nerves lie alongside or penetrate the bone.
- Detailed examination of these bone markings, or surface features, can yield an abundance of anatomical information.
- For example, anthropologists, criminologists, and pathologists can often determine the size, age, sex, and general appearance of an individual on the basis of incomplete skeletal remains.

**Figure 6-2** An Introduction to Bone Markings.



**Table 6.1 Bone Markings**

| NAME OF BONE MARKING                                                | DESCRIPTION                                                                        | ILLUSTRATIONS                                                                       |
|---------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Projections That Are Sites of Muscle and Ligament Attachment</b> |                                                                                    |                                                                                     |
| Tuberosity (too"bĕ-ros"ĭ-te)                                        | Large rounded projection; may be roughened                                         |   |
| Crest                                                               | Narrow ridge of bone; usually prominent                                            |                                                                                     |
| Trochanter (tro-kan'ter)                                            | Very large, blunt, irregularly shaped process (the only examples are on the femur) |                                                                                     |
| Line                                                                | Narrow ridge of bone; less prominent than a crest                                  |                                                                                     |
| Tubercle (too'ber-kl)                                               | Small rounded projection or process                                                |                                                                                     |
| Epicondyle (ep"ĭ-kon'dĭl)                                           | Raised area on or above a condyle                                                  |                                                                                     |
| Spine                                                               | Sharp, slender, often pointed projection                                           |                                                                                     |
| Process                                                             | Any bony prominence                                                                |                                                                                     |
| <b>Projections That Help to Form Joints</b>                         |                                                                                    |                                                                                     |
| Head                                                                | Bony expansion carried on a narrow neck                                            |   |
| Facet                                                               | Smooth, nearly flat articular surface                                              |                                                                                     |
| Condyle (kon'dĭl)                                                   | Rounded articular projection                                                       |                                                                                     |
| Ramus (ra'mus)                                                      | Armlike bar of bone                                                                |                                                                                     |
| <b>Depressions and Openings</b>                                     |                                                                                    |                                                                                     |
| <b>For Passage of Blood Vessels and Nerves</b>                      |                                                                                    |                                                                                     |
| Groove                                                              | Furrow                                                                             |  |
| Fissure                                                             | Narrow, slitlike opening                                                           |                                                                                     |
| Foramen (fo-ra'men)                                                 | Round or oval opening through a bone                                               |                                                                                     |
| Notch                                                               | Indentation at the edge of a structure                                             |                                                                                     |
| <b>Others</b>                                                       |                                                                                    |                                                                                     |
| Meatus (me-a'tus)                                                   | Canal-like passageway                                                              |                                                                                     |
| Sinus                                                               | Cavity within a bone, filled with air and lined with mucous membrane               |                                                                                     |
| Fossa (fos'ah)                                                      | Shallow, basinlike depression in a bone, often serving as an articular surface     |                                                                                     |

# BLOOD AND NERVE SUPPLY OF BONE

- Bone is supplied with blood by:
  - ▣ **Periosteal arteries** accompanied by nerves supply the periosteum and compact bone
  - ▣ **Epiphyseal veins** carry blood away from long bones
- Nerves accompany the blood vessels that supply bones
  - ▣ The periosteum is rich in sensory nerves sensitive to tearing or tension

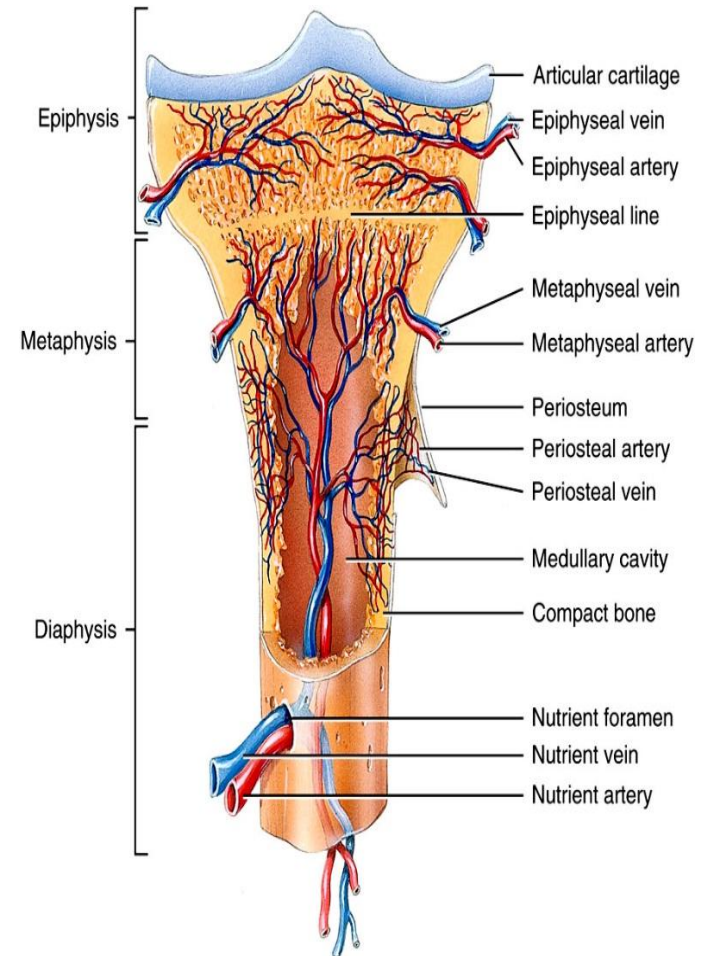
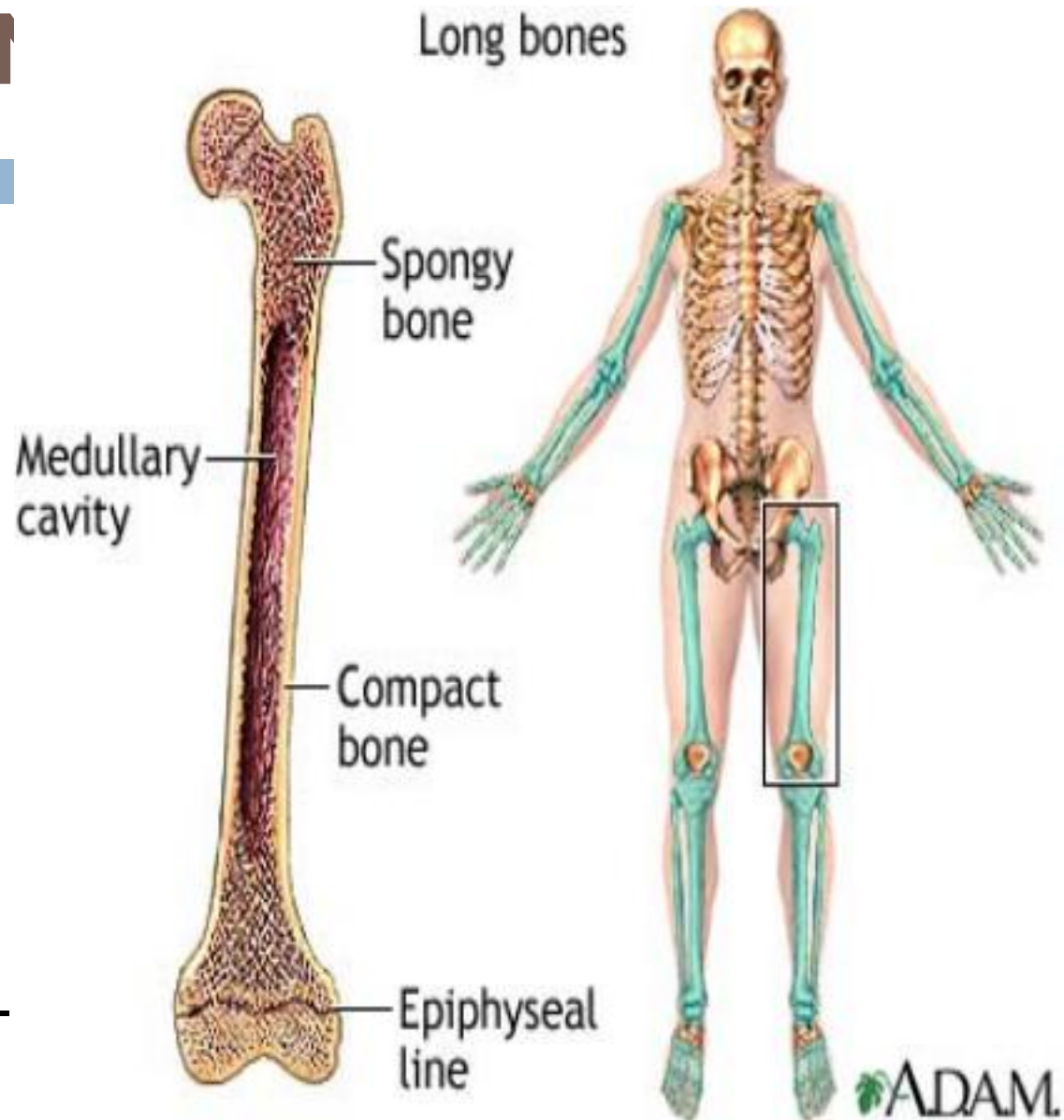


Figure 06.04 Tortora - PAP 12/e  
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# LONG BONE

**Compact Bone** –  
dense outer layer

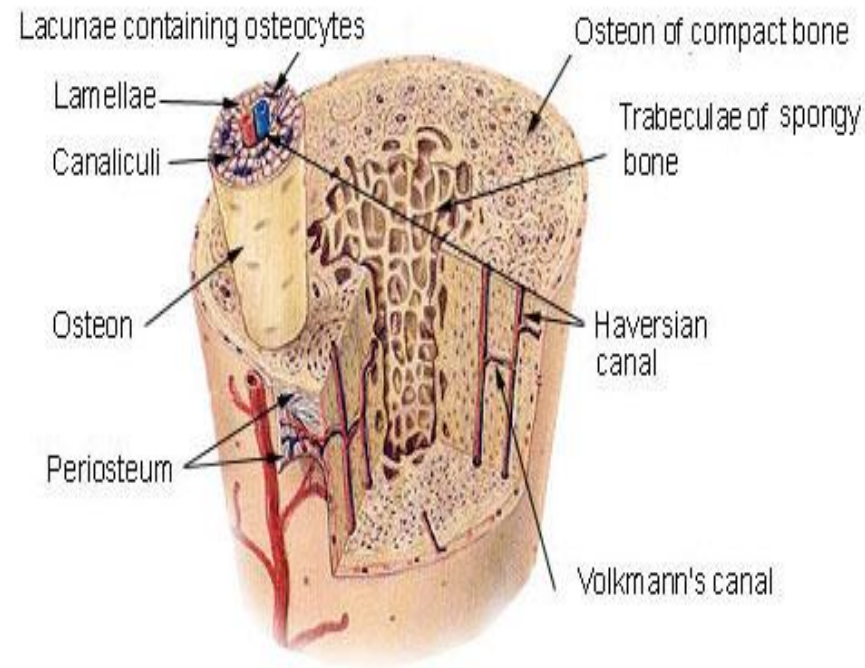
**Spongy Bone** –  
(cancellous bone)  
honeycomb of  
trabeculae (needle-  
like or flat pieces)  
filled with bone  
marrow



# COMPACT BONE: (OSTEON) EXTERNAL LAYER

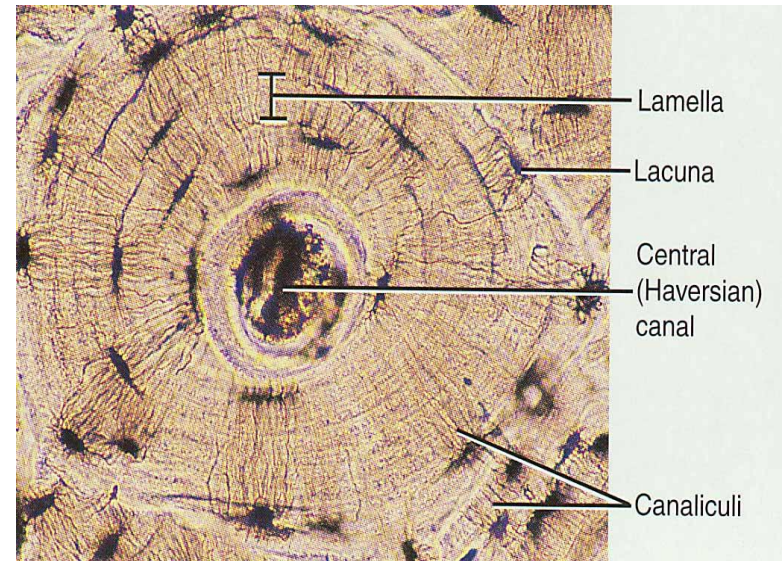
- called **lamellar bone** (groups of elongated tubules called lamella)
- majority of all long bones
- protection and strength (wt. bearing)
- concentric ring structure
- blood vessels and nerves penetrate periosteum through horizontal openings called perforating (Volkmann's) canals.

**Compact Bone & Spongy (Cancellous Bone)**



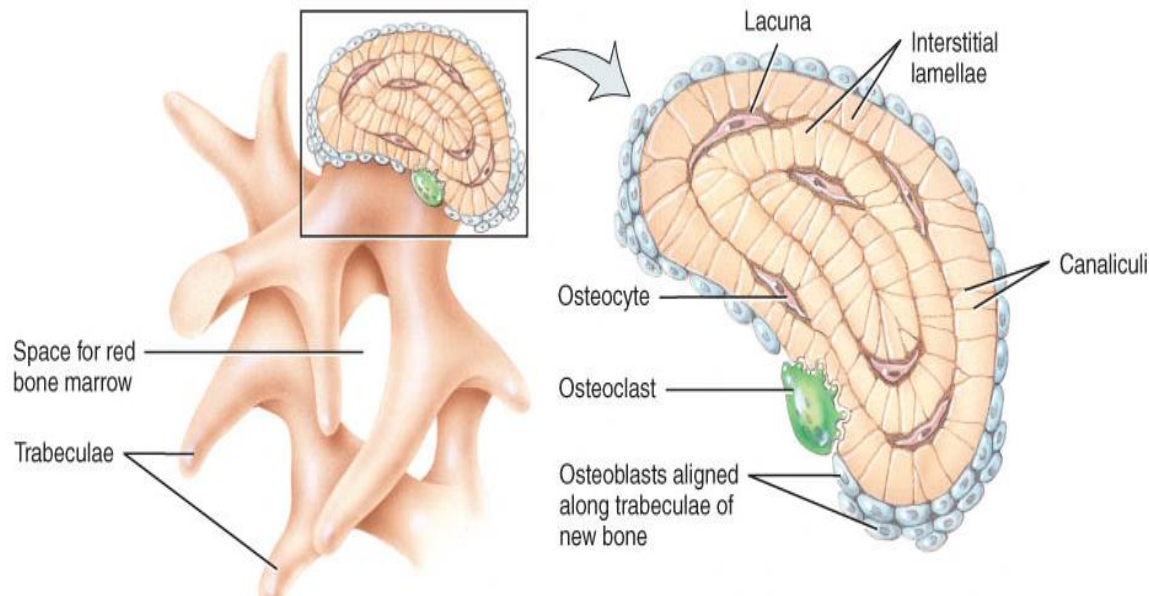
# COMPACT BONE: (OSTEON) EXTERNAL LAYER

- **Central (Haversian) canals** run longitudinally. Blood vessels and nerves.
  - around canals are concentric lamella
  - osteocytes occupy lacunae which are between the lamella
  - radiating from the lacunae are channels called canaliculi (finger like processes of osteocytes)



# SPONGY BONE (CANCELOUS BONE): INTERNAL LAYER

- trabecular bone tissue (haphazard arrangement).
- filled with red and yellow bone marrow
- osteocytes get nutrients directly from circulating blood.
- short, flat and irregular bone is made up of mostly spongy bone



# HISTOLOGY OF BONE - MATRIX

## Matrix

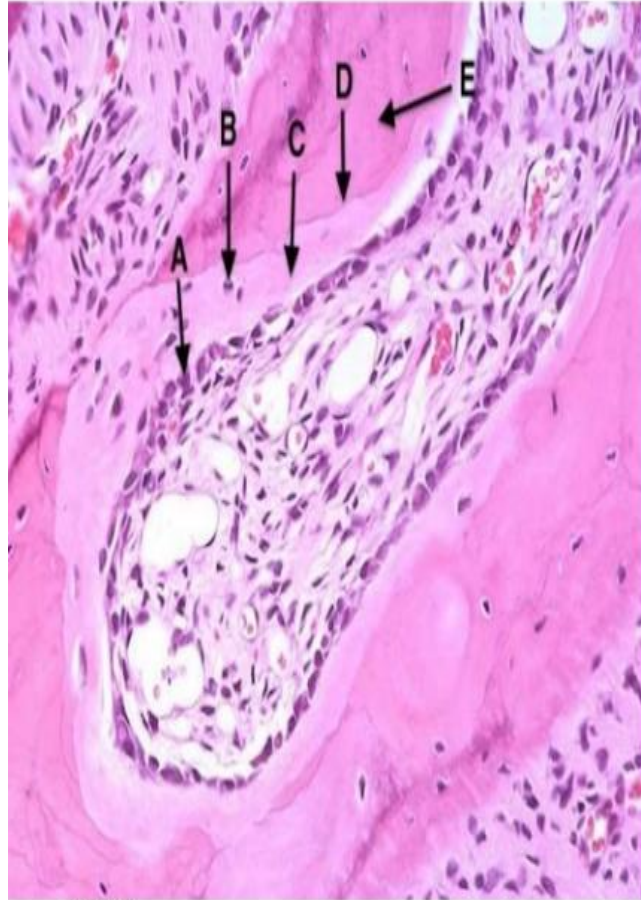
-Cells are surrounded by matrix.

- ❖ 25% water
- ❖ 25% protein
- ❖ 50% mineral salts

- Matrix has organic and inorganic component

- ❖ Organic (non-calcified) – type I collagen + non-mineralized ground substance (chondroitin sulfate and keratin sulfate), found in “new bone” or osteoid

- ❖ Inorganic (calcified) – hydroxyapatite (calcium and phosphorus), mineralized calcified osteoid



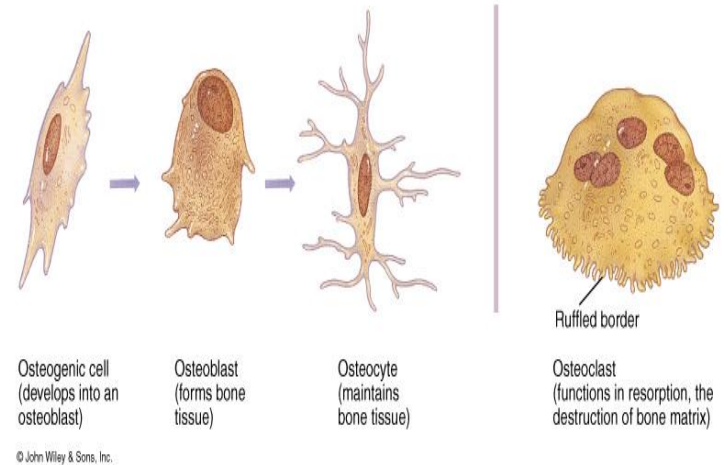
- A = osteoblasts,
- B = osteocytes,
- C = osteoid,
- D = cement line
- E = bone

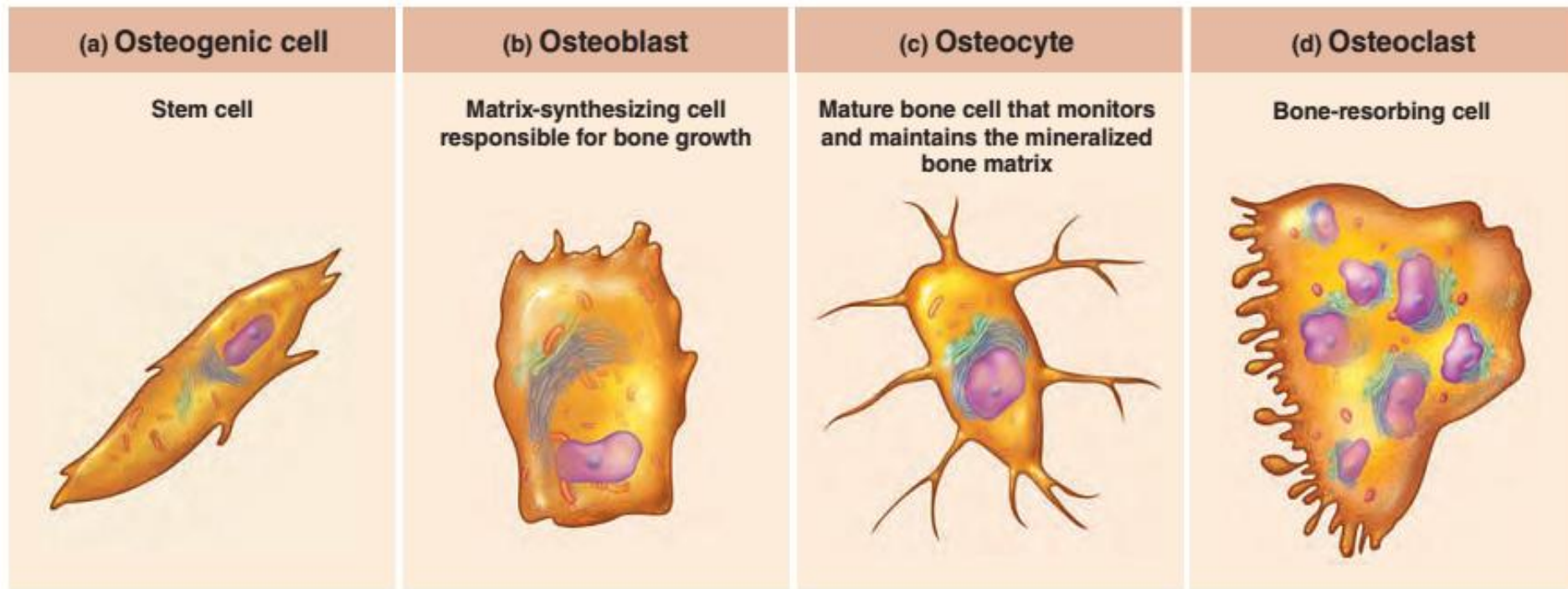
# HISTOLOGY OF BONE - CELLS OF THE BONE

□ 4 cell types make up osseous tissue:

- Osteoprogenitor cells
- Osteoblasts
- Osteocytes
- Osteoclasts

## Cells of Bone Tissue



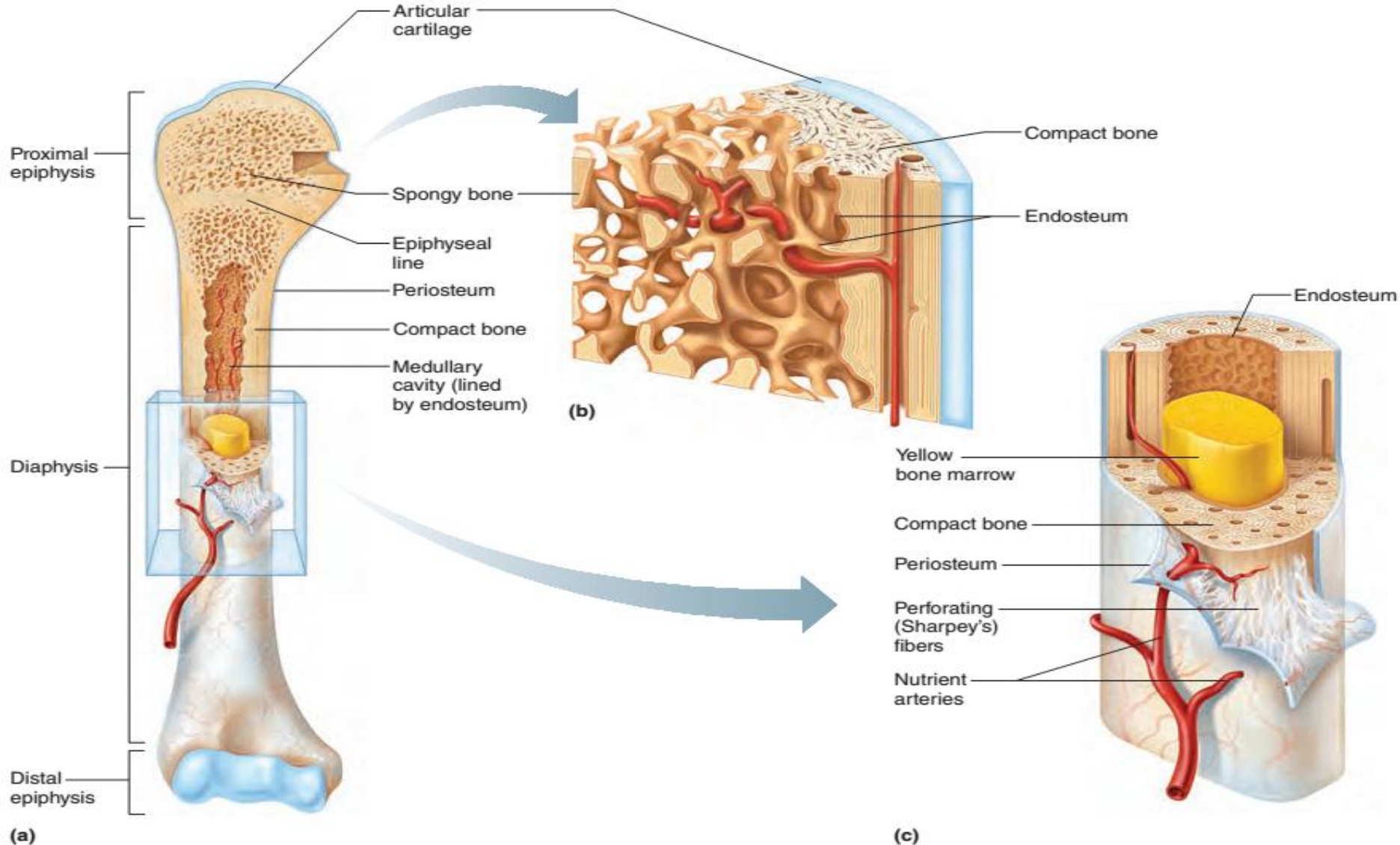


**Figure 6.5** Comparison of different types of bone cells. The bone lining cell, similar in appearance to the osteogenic cell and similar to the osteocyte in function, is not illustrated.

**Osteogenic Cells** Osteogenic cells, also called osteoprogenitor cells, are mitotically active stem cells found in the membranous periosteum and endosteum.

In growing bones they are flattened or squamous cells.

When stimulated, these cells differentiate into osteoblasts or bone lining cells, while others persist as osteogenic cells.



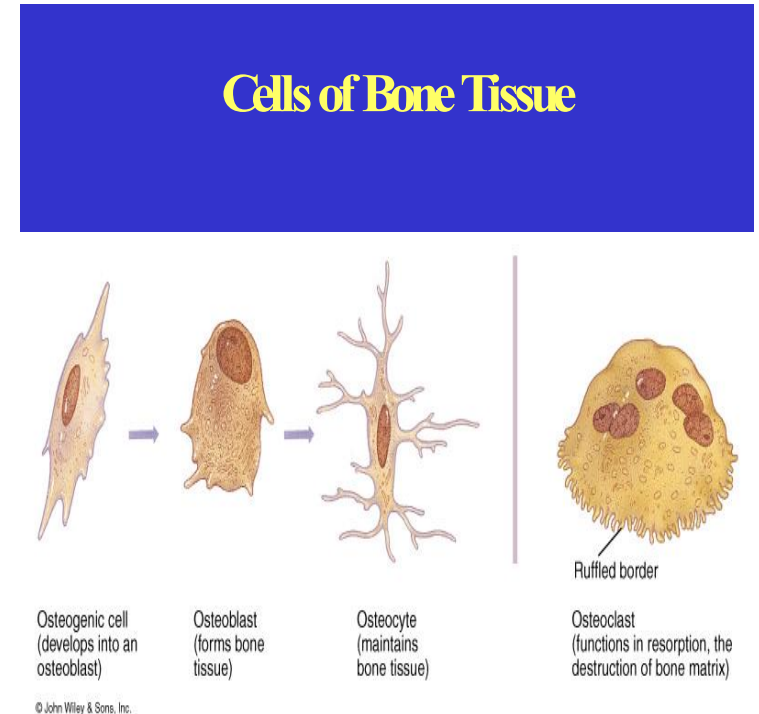
**Figure 6.4 The structure of a long bone (humerus of arm).** (a) Anterior view with bone sectioned frontally to show the interior at the proximal end. (b) Enlarged

view of spongy bone and compact bone of the epiphysis of (a). (For related images, see *A Brief Atlas of the Human Body*, Plates 20 and 21.) (c) Enlarged cross-sectional view

of the shaft (diaphysis) of (a). Note that the external surface of the diaphysis is covered by periosteum, but the articular surface of the epiphysis is covered with hyaline cartilage.

# CELLS OF THE BONE

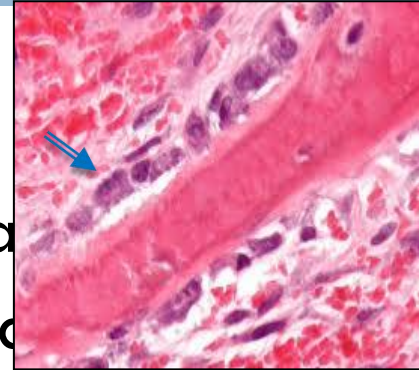
- **Osteoprogenitor cells:**
  - derived from mesenchyme
  - all connective tissue is derived
  - unspecialized stem cells
  - undergo mitosis and develop into osteoblasts
  - found on inner surface of periosteum and endosteum.



# CELLS OF THE BONE

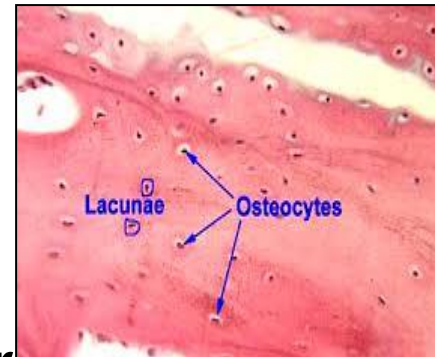
- **Osteoblasts:**

- bone forming cells
- found on surface of bone (on endosteum and on internal surface of trabeculae)
- no ability to mitotically divide
- collagen secretors



- **Osteocytes:**

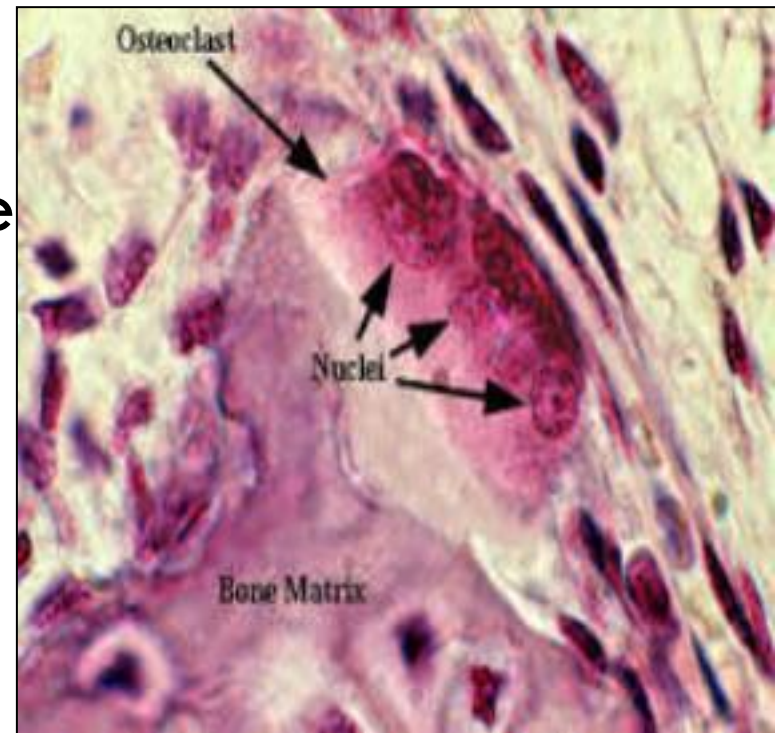
- mature bone cells
- derived from osteoblasts
- do not secrete matrix material
- cellular duties include exchange of nutrients and waste with blood.



# CELLS OF THE BONE

## □ Osteoclasts

- bone resorbing cells
- bone surface
- growth, maintenance and bone

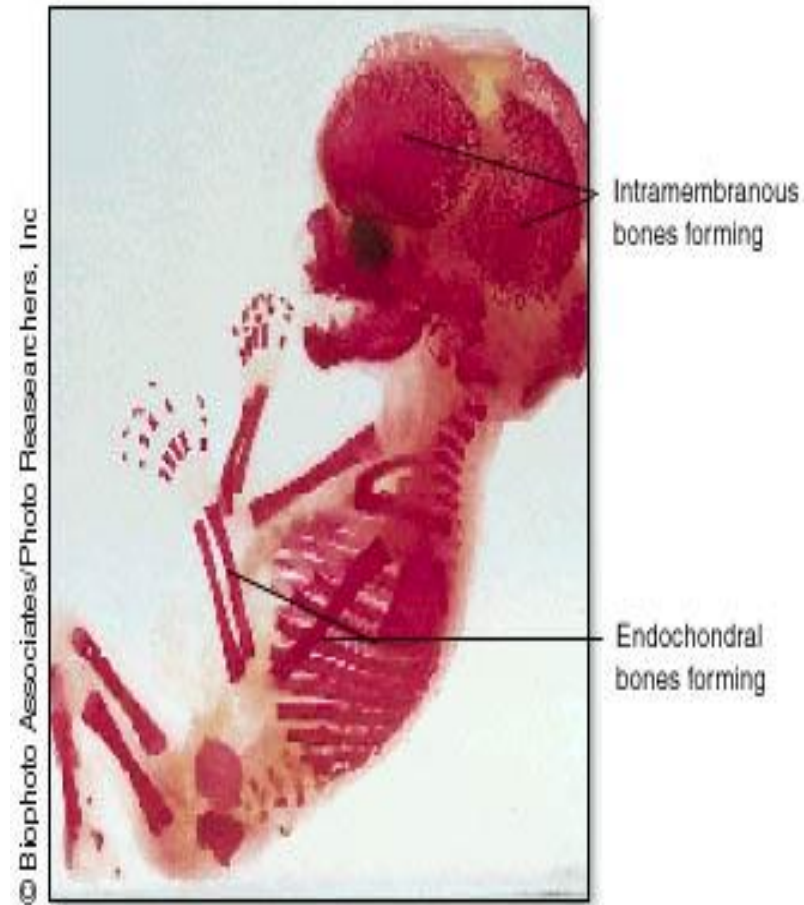


# BONE FORMATION

- The process of bone formation is called **ossification/osteogenesis**
- Bone formation occurs in four situations:
  - ▣ 1) Formation of bone in an embryo
  - ▣ 2) Growth of bones until adulthood
  - ▣ 3) Remodeling of bone
  - ▣ 4) Repair of fractures

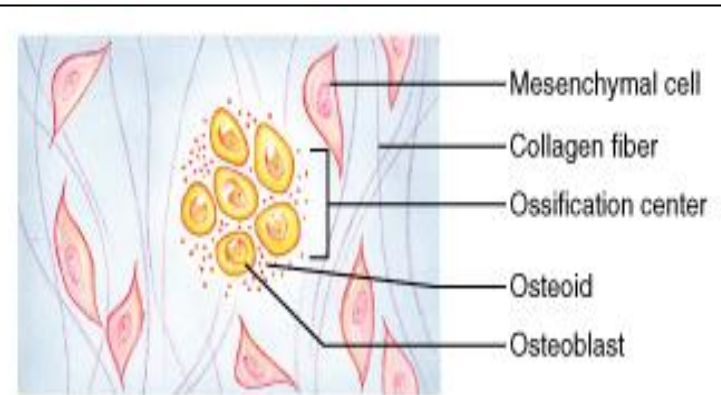
# BONE FORMATION

- **Formation of Bone in an Embryo**
  - cartilage formation and ossification occurs during the sixth week of embryonic development
  - two patterns
    - **Intramembranous ossification**
      - Flat bones of the skull and mandible are formed in this way
      - “Soft spots” that help the fetal skull pass through the birth canal later become ossified forming the skull
    - **Endochondral ossification**
      - The replacement of cartilage by bone
      - Most bones of the body are formed in this way including long bones



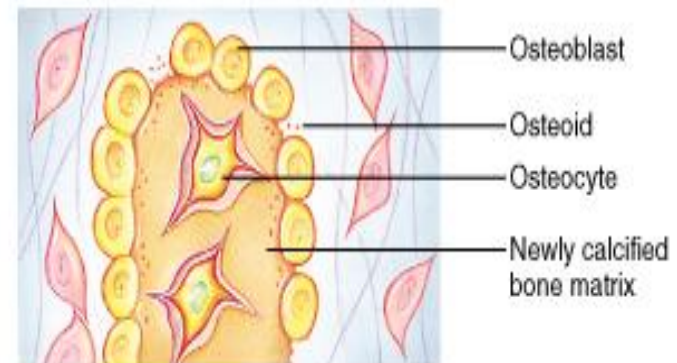
# INTRAMEMBRANOUS OSSIFICATION

- An ossification center appears in the fibrous connective tissue membrane
- Osteoblasts secrete bone matrix within the fibrous membrane
- Osteoblasts mature into osteocytes



- ① An ossification center appears in the fibrous connective tissue membrane.

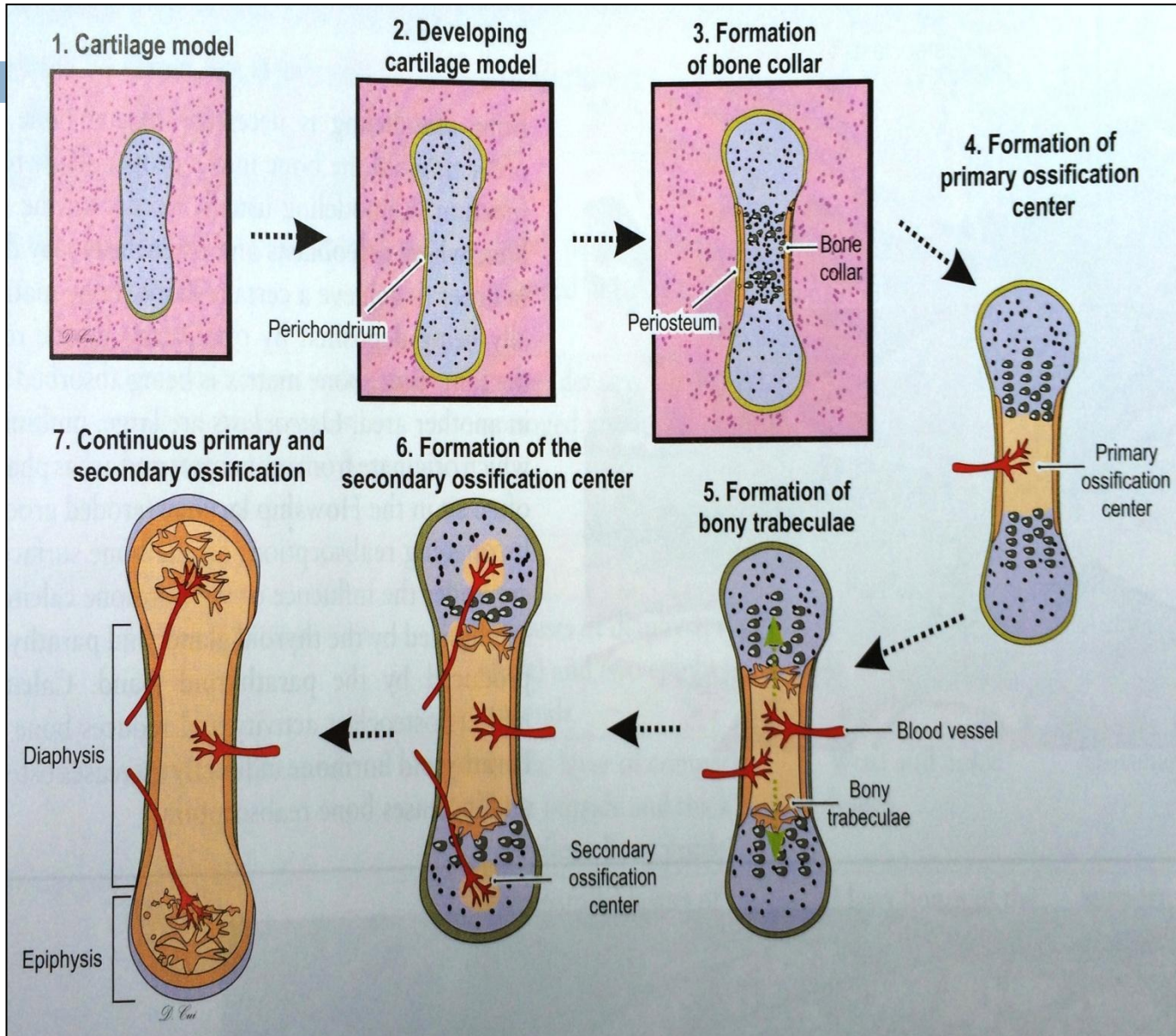
- Selected centrally located mesenchymal cells cluster and differentiate into osteoblasts, forming an ossification center.



- ② Bone matrix (osteoid) is secreted within the fibrous membrane.

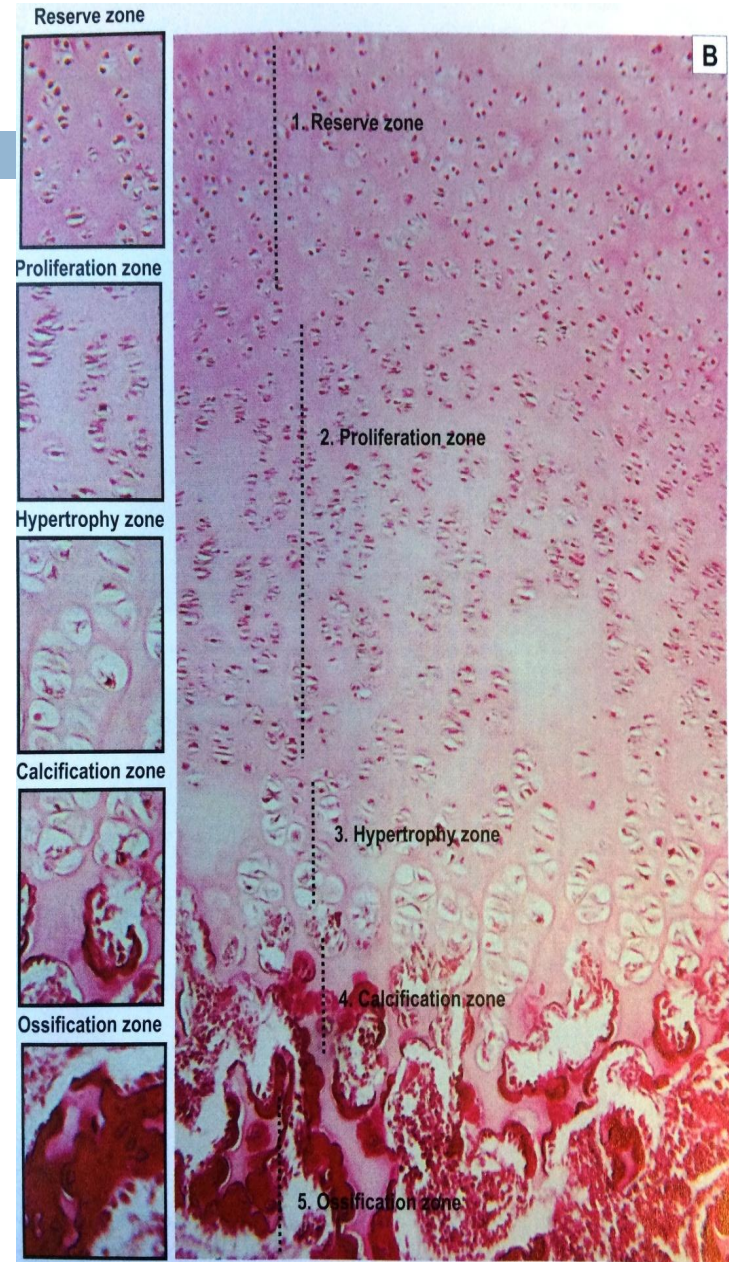
- Osteoblasts begin to secrete osteoid, which is mineralized within a few days.
- Trapped osteoblasts become osteocytes.

# ENDOCHONDRAL OSSIFICATION



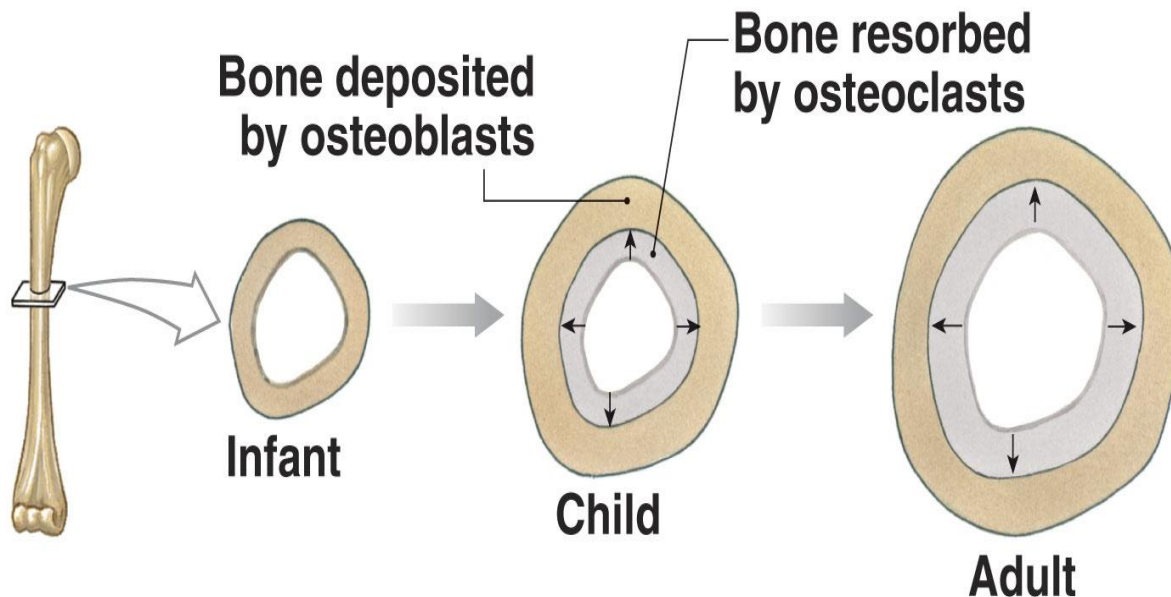
# BONE GROWTH AT EPIPHYSEAL PLATE

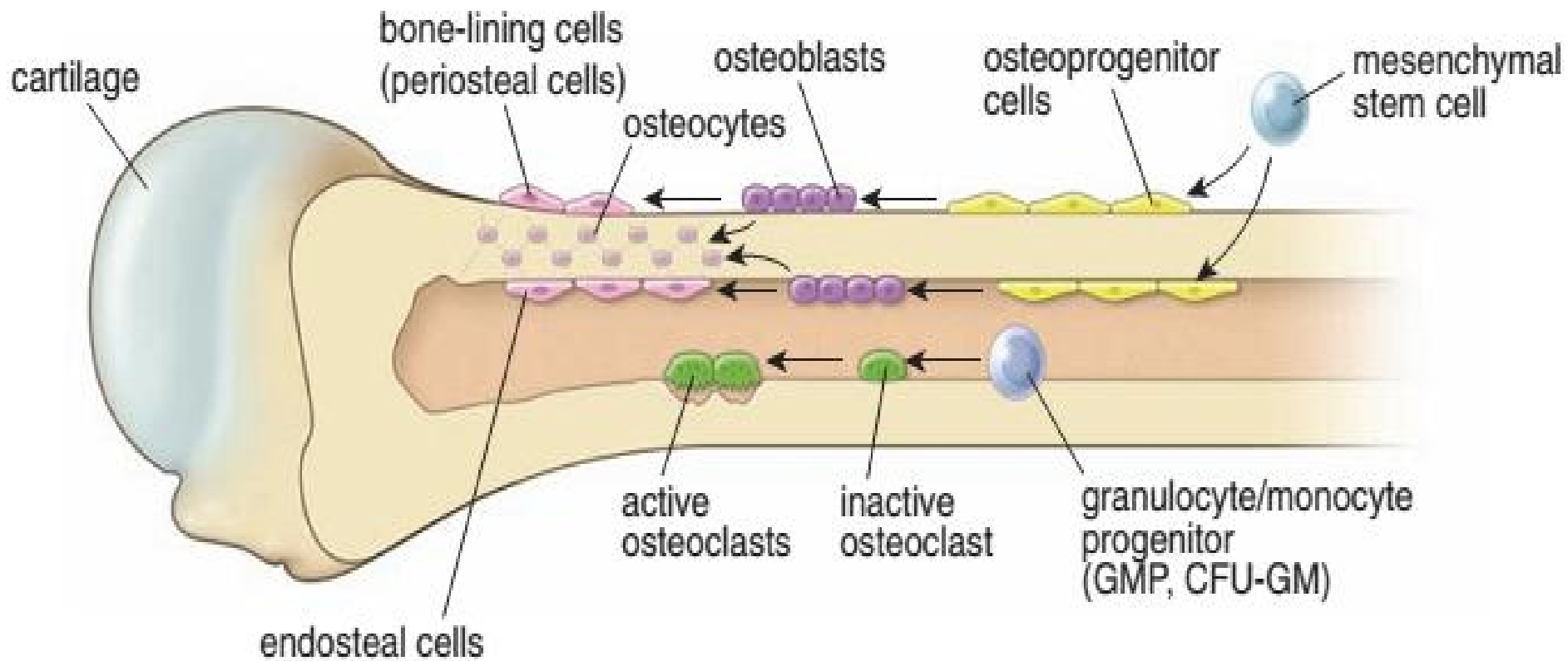
- Reserve zone – inactive chondrocytes not arranged in groups
- Proliferation zone – chondrocytes undergo mitosis and arranged in columns
- Hypertrophy zone – mature chondrocytes with increased size
- Calcification zone – cartilage matrix becomes calcified
- Ossification zone – blood vessels and primary marrow, bone deposition



# APPOSITIONAL BONE GROWTH

- Growing bones widen as they lengthen
- Appositional growth – growth of a bone by addition of bone tissue to its surface
- Bone is resorbed at endosteal surface and added at periosteal surface
  - ▣ Osteoblasts – add bone tissue to the external surface of the diaphysis
  - ▣ Osteoclasts – remove bone from the internal surface of the diaphysis





Schematic drawing of cells associated with bone.

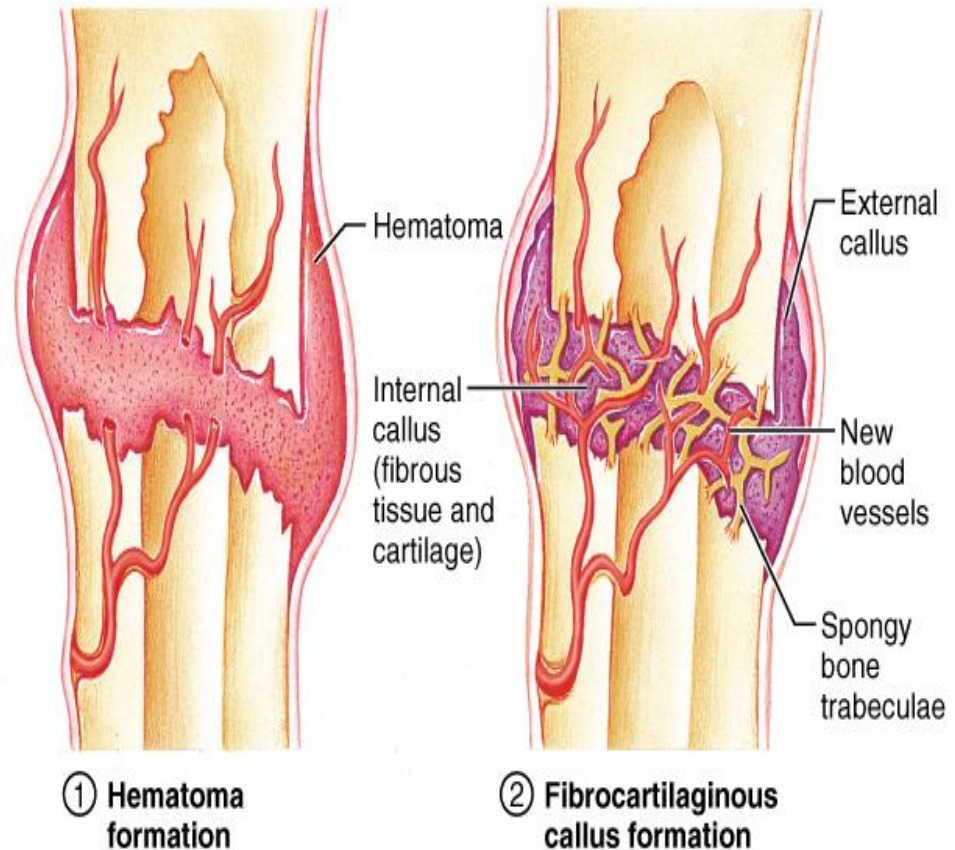
# BONE REMODELING

- bone continually renews itself
- never metabolically at rest
- enables Ca to be pulled from bone when blood levels are low
- osteoclasts are responsible for matrix destruction
- produce lysosomal enzymes and acids
- spongy bone replaced every 3-4 years
- compact bone every 10 years

# BONE REMODELING

- **Fractures:** Any bone break.

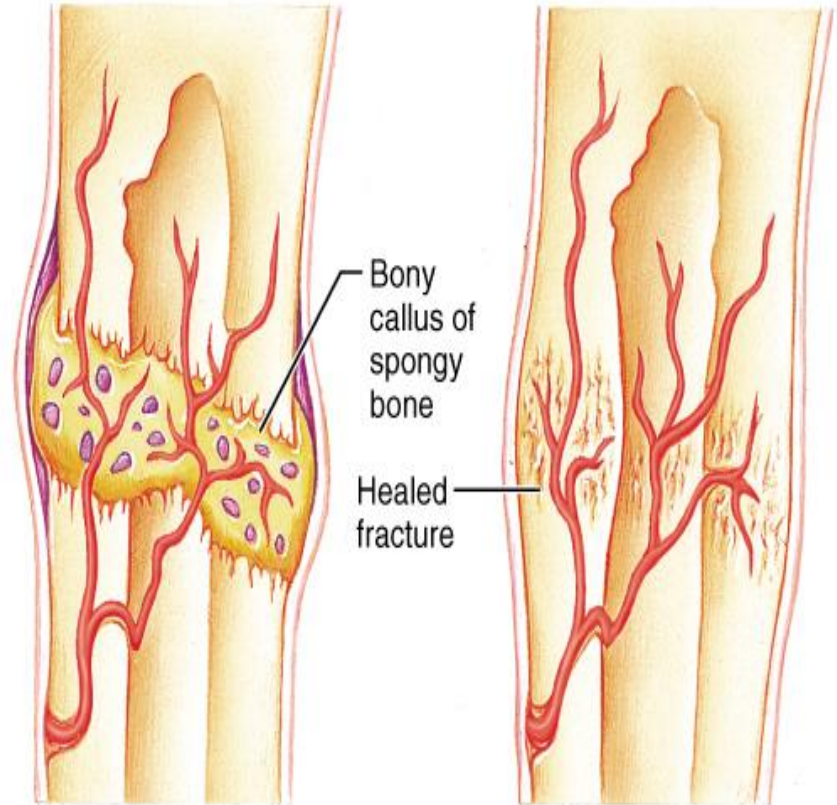
- blood clot will form around break
- fracture hematoma
- inflammatory process begins
- blood capillaries grow into clot
- phagocytes and osteoclasts remove damaged tissue
- procallus forms and is invaded by osteoprogenitor cells and fibroblasts
- collagen and fibrocartilage turns procallus to fibrocartilaginous (soft) callus



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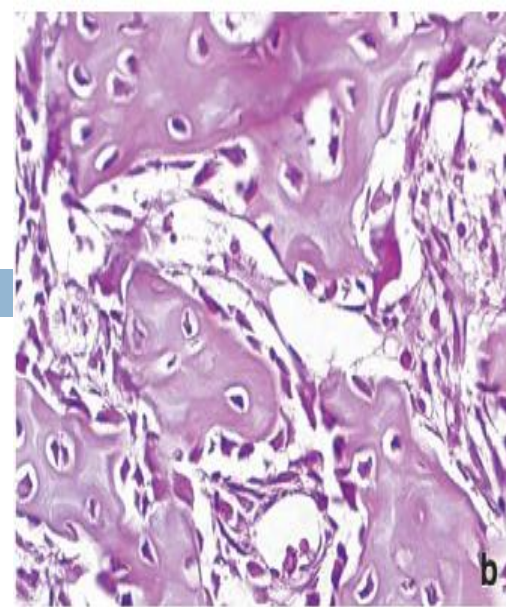
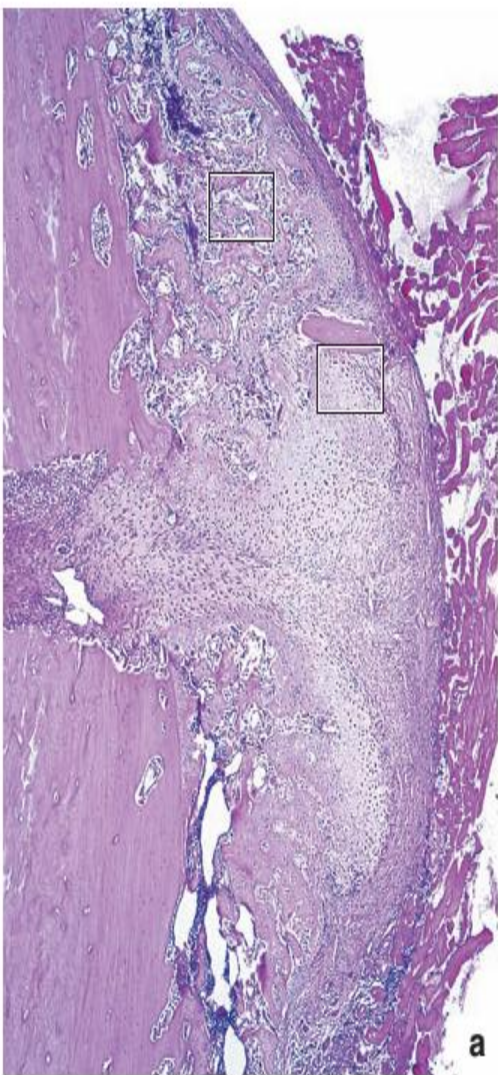
# BONE REMODELING

- ❑ Broken ends of bone are bridged by callus
- ❑ osteoprogenitor cells are replaced by osteoblasts and form spongy bone
- ❑ bony (hard) callus is formed
- ❑ callus is resorbed by osteoclasts and compact bone replaces spongy bone.
- ❑ **Remodeling** : the shaft is reconstructed to resemble original unbroken bone.
- ❑ **Closed reduction** - bone ends coaxed back into place by manipulation
- ❑ **Open reduction** - surgery, bone ends secured together with pins or wires



③ Bony callus formation

④ Bone remodeling



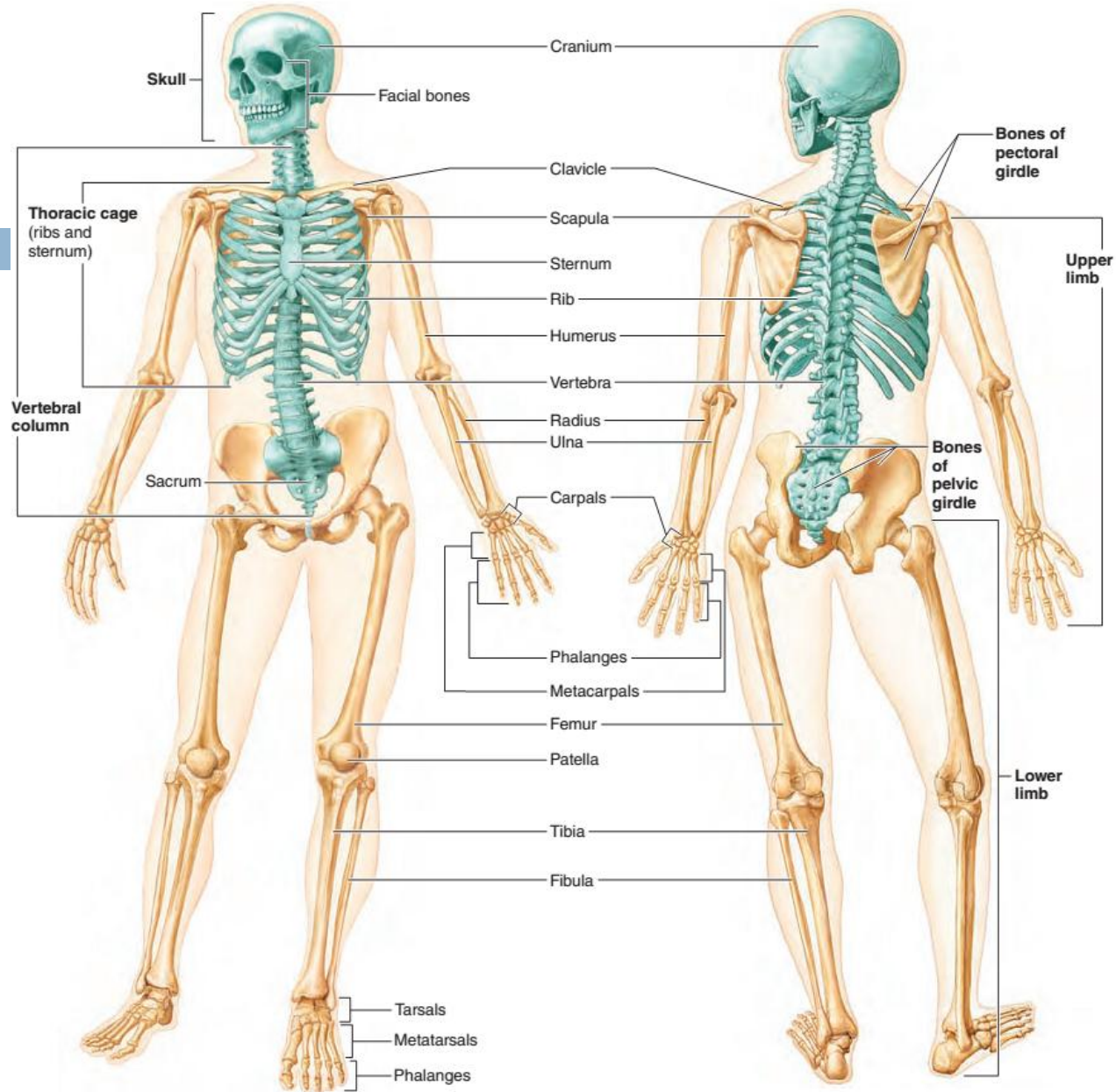
x300.



x300.

Photomicrograph of fractured long bone undergoing repair.

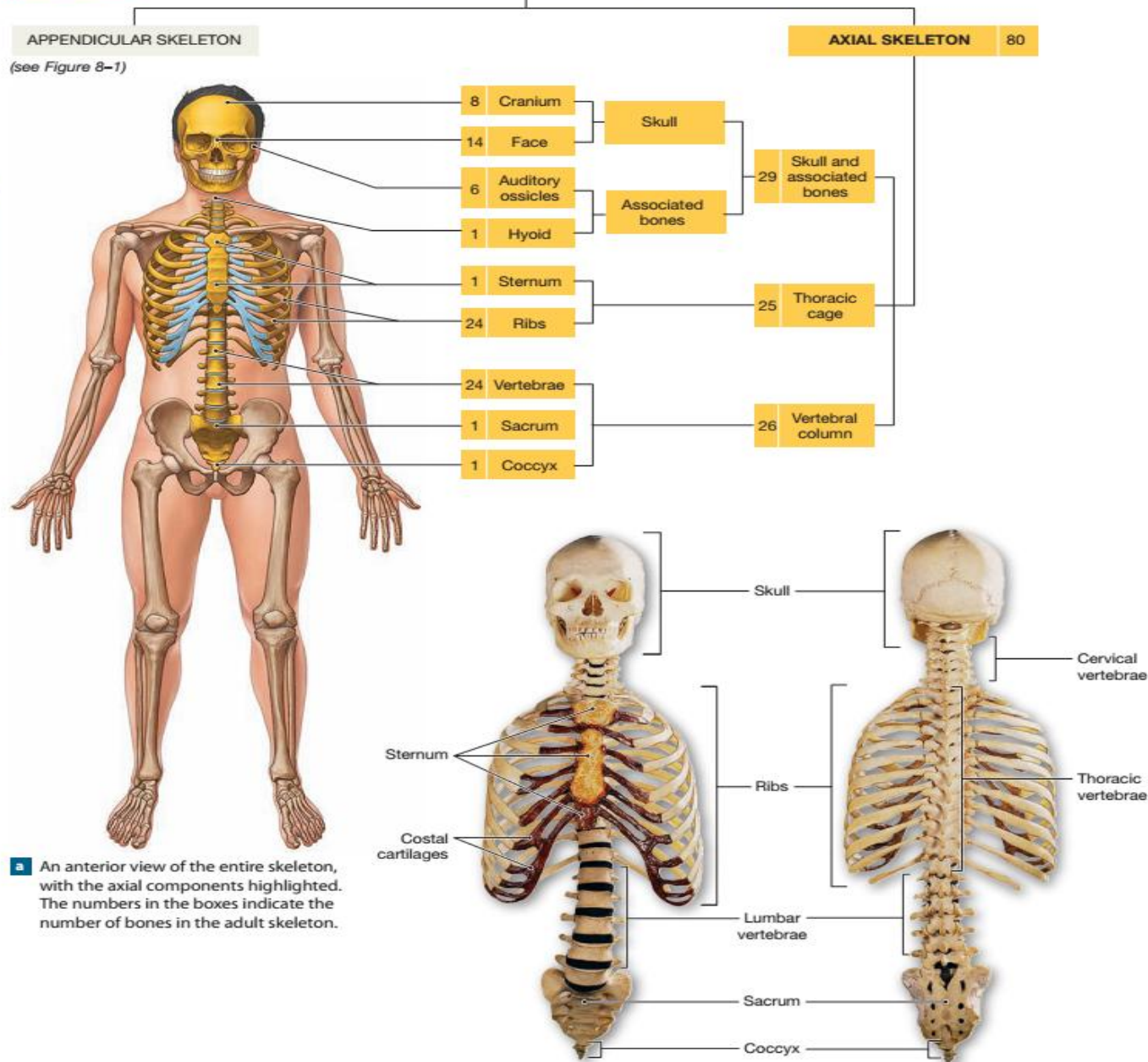
a. This low-magnification photomicrograph of a 3-week-old bone fracture, stained with H&E, x35.



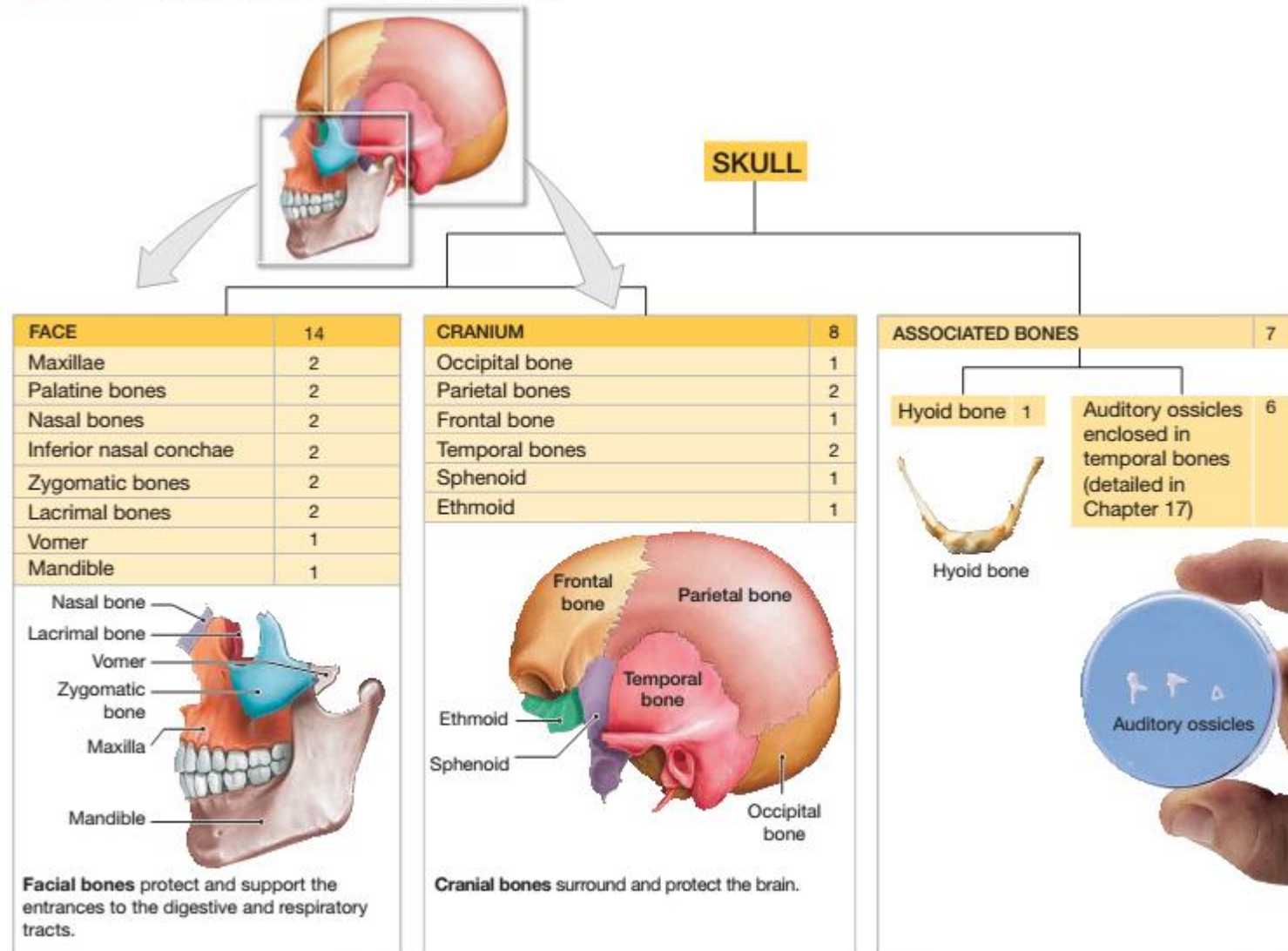
(a) Anterior view

(b) Posterior view

**Figure 7.1 The human skeleton.** Bones of the axial skeleton are colored green. Bones of the appendicular skeleton are gold.



**Figure 7-2** Cranial and Facial Subdivisions of the Skull.



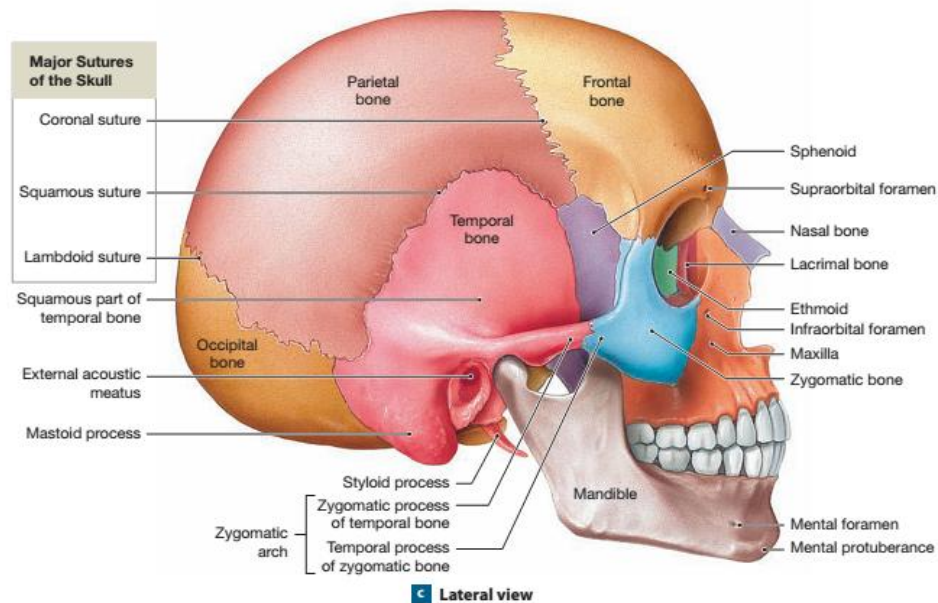
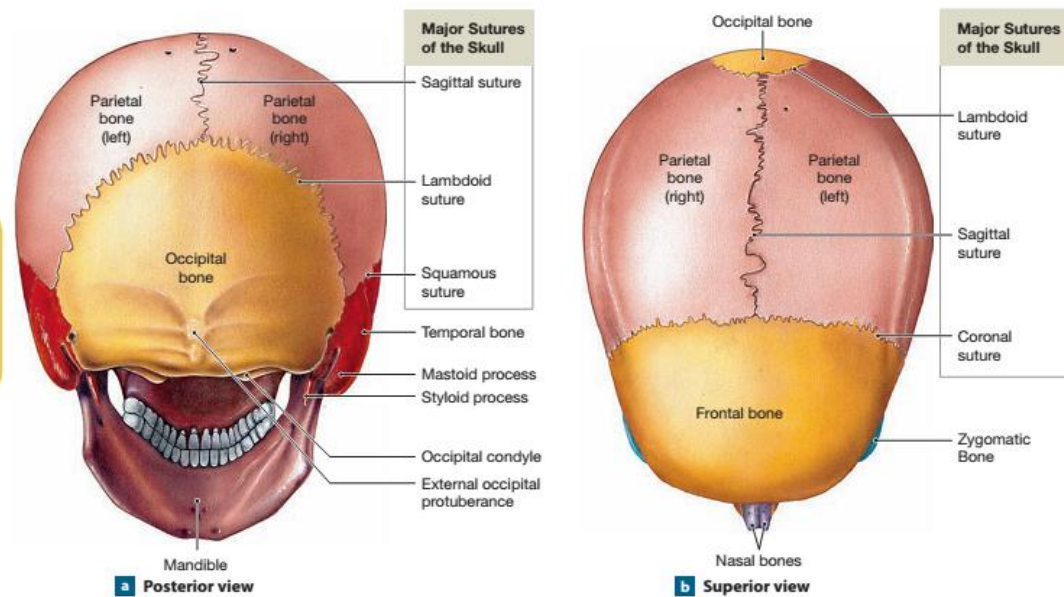
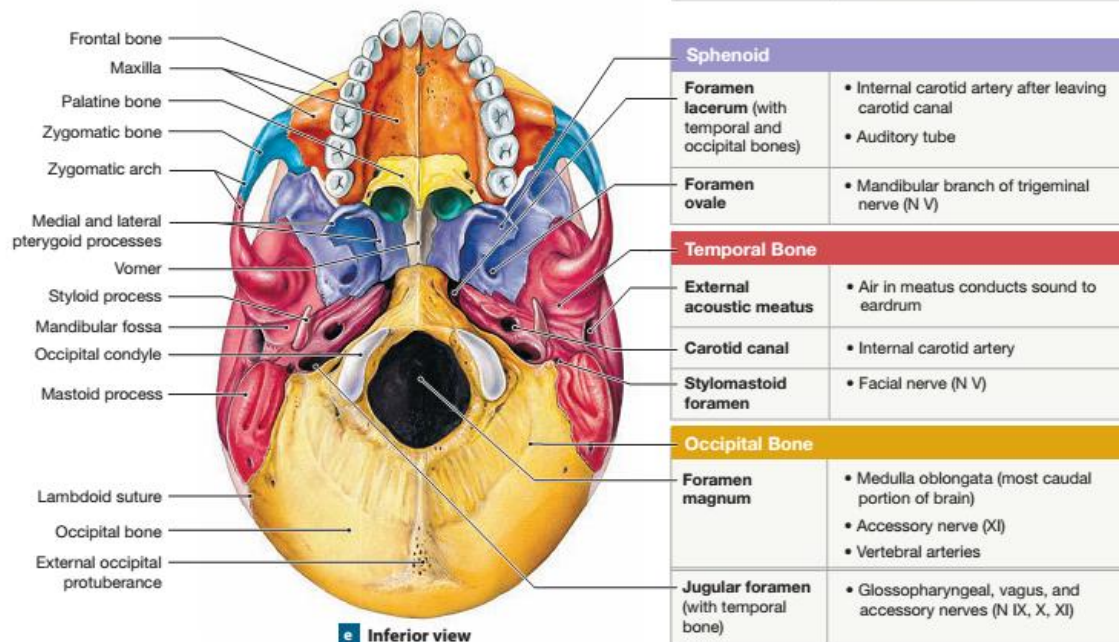
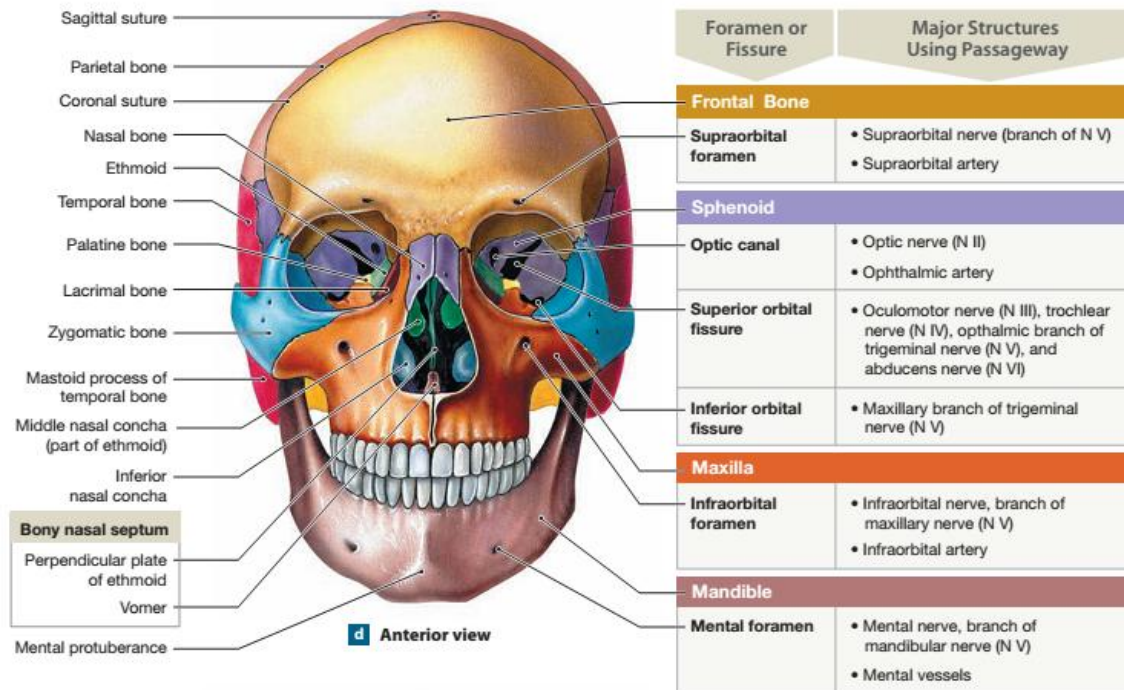
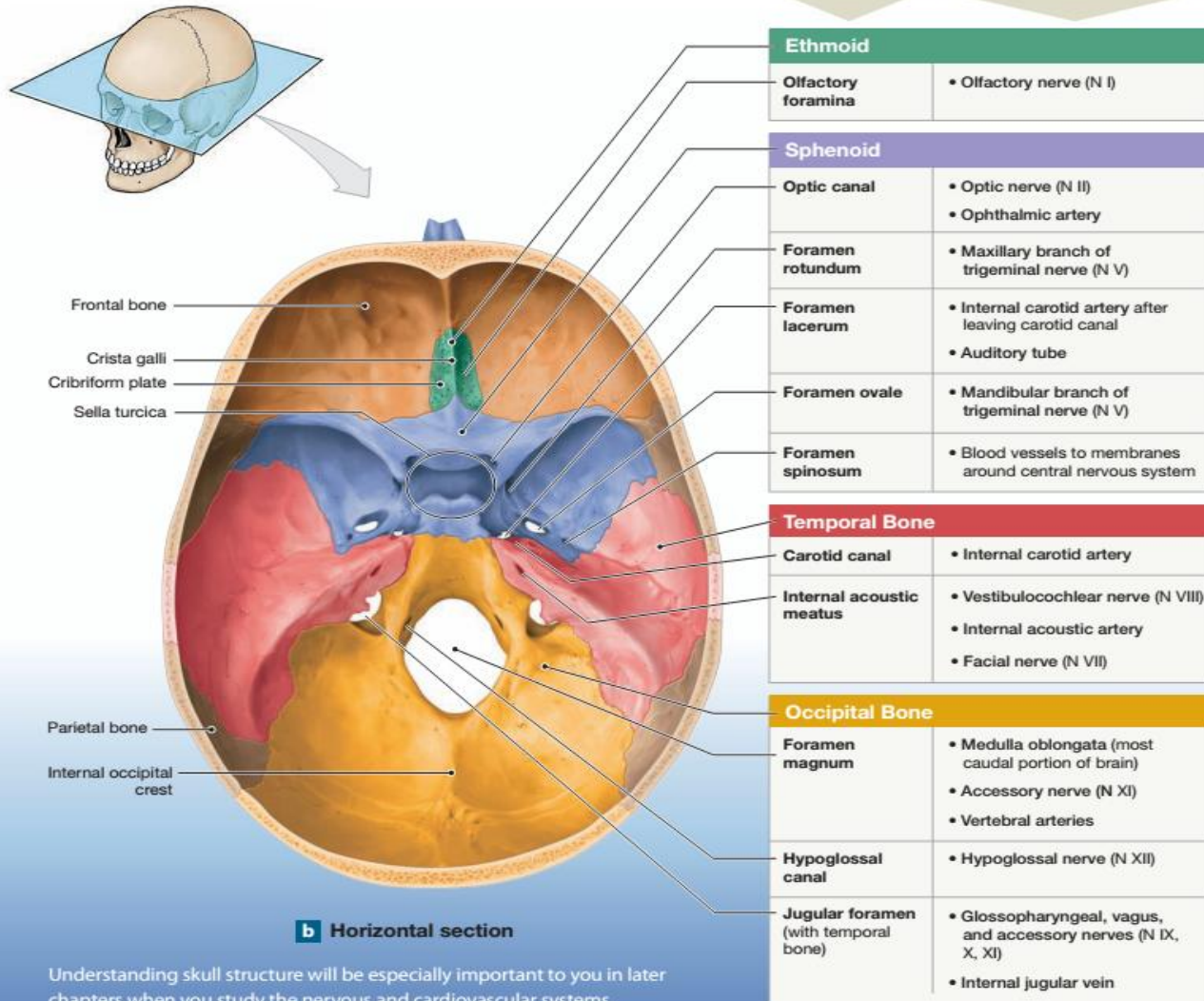


Figure 7-3 The Adult Skull (continued).



## THE FLOOR OF THE CRANIUM

This superior view of a horizontal section through the skull shows the floor of the cranial cavity. This is where most of the nerves and blood vessels enter or leave the cranial cavity. Compare this figure and chart with *Figure 7–3e* showing the inferior view of the skull and *Figure 14–19* (p. 492) showing the origin of the cranial nerves.



Understanding skull structure will be especially important to you in later chapters when you study the nervous and cardiovascular systems.

**Figure 7-17** The Vertebral Column. *ATLAS: Plate 2b*

### Spinal Curves

Primary curves develop before birth, and secondary curves after birth.

The **cervical curve**, a secondary curve, develops as the infant learns to balance the weight of the head on the vertebrae of the neck.

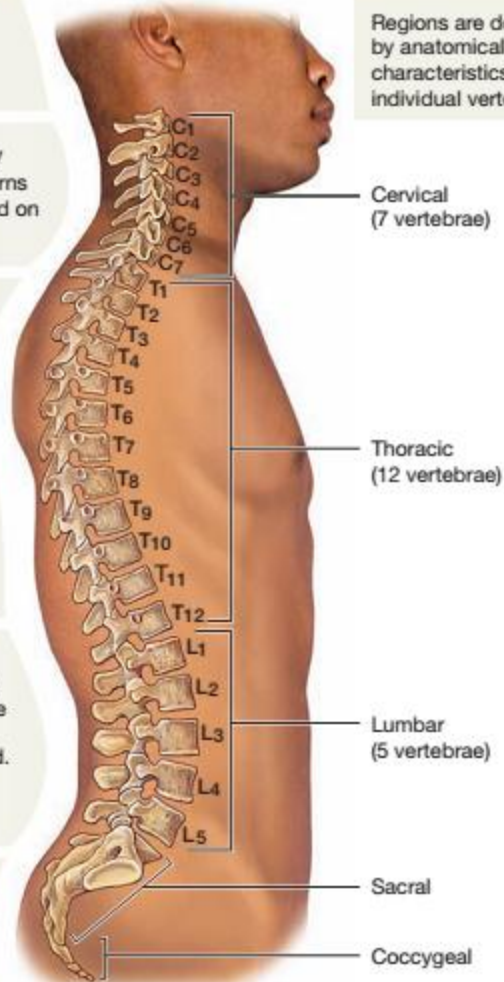
The **thoracic curve**, a primary curve, accommodates the thoracic organs.

The **lumbar curve**, a secondary curve, balances the weight of the trunk over the lower limbs; it develops with the ability to stand.

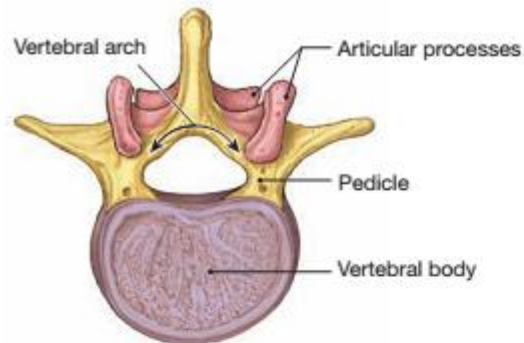
The **sacral curve**, a primary curve, accommodates the abdominopelvic organs.

### Vertebral Regions

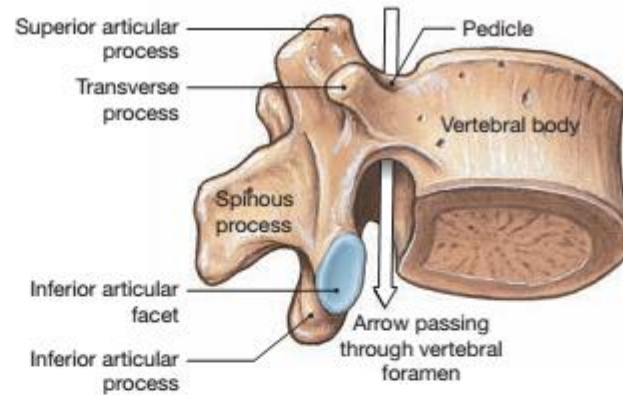
Regions are defined by anatomical characteristics of individual vertebrae.



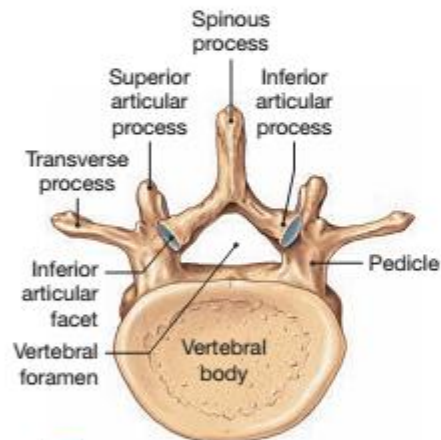
**Figure 7-18** Vertebral Anatomy.



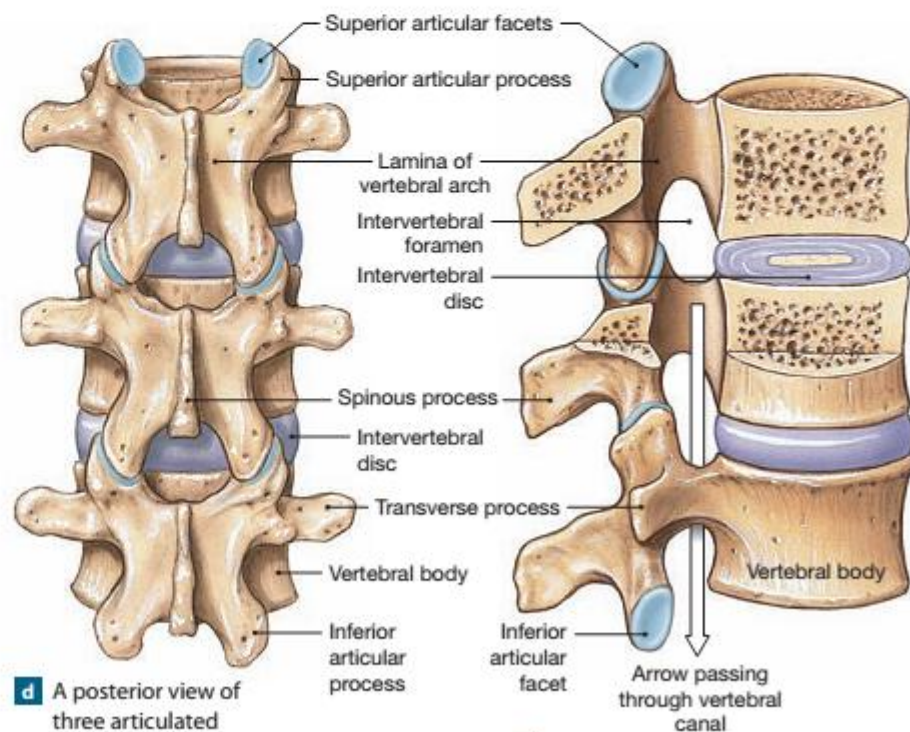
**a** The major components of a typical vertebra



**b** A lateral and slightly inferior view of a vertebra



**c** An inferior view of a vertebra



**d** A posterior view of three articulated vertebrae

**e** A lateral and sectional view of three articulated vertebrae



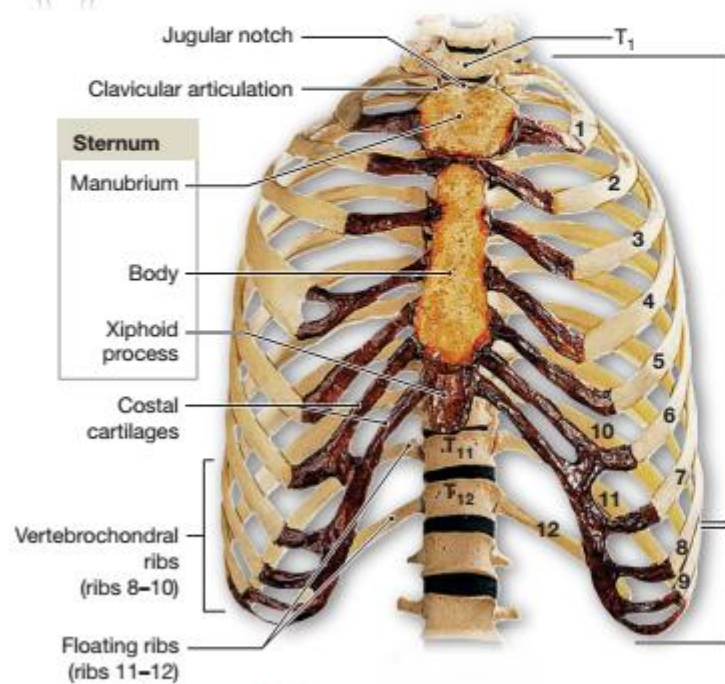
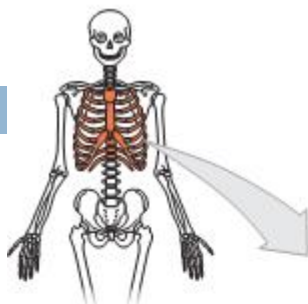
**a** Kyphosis



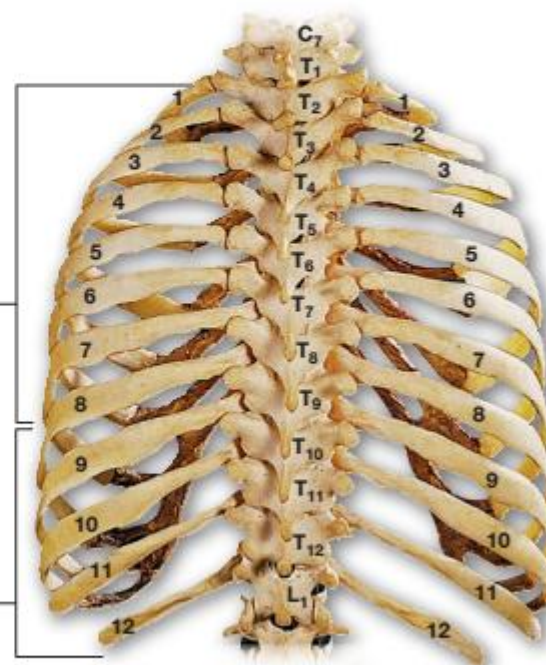
**b** Lordosis



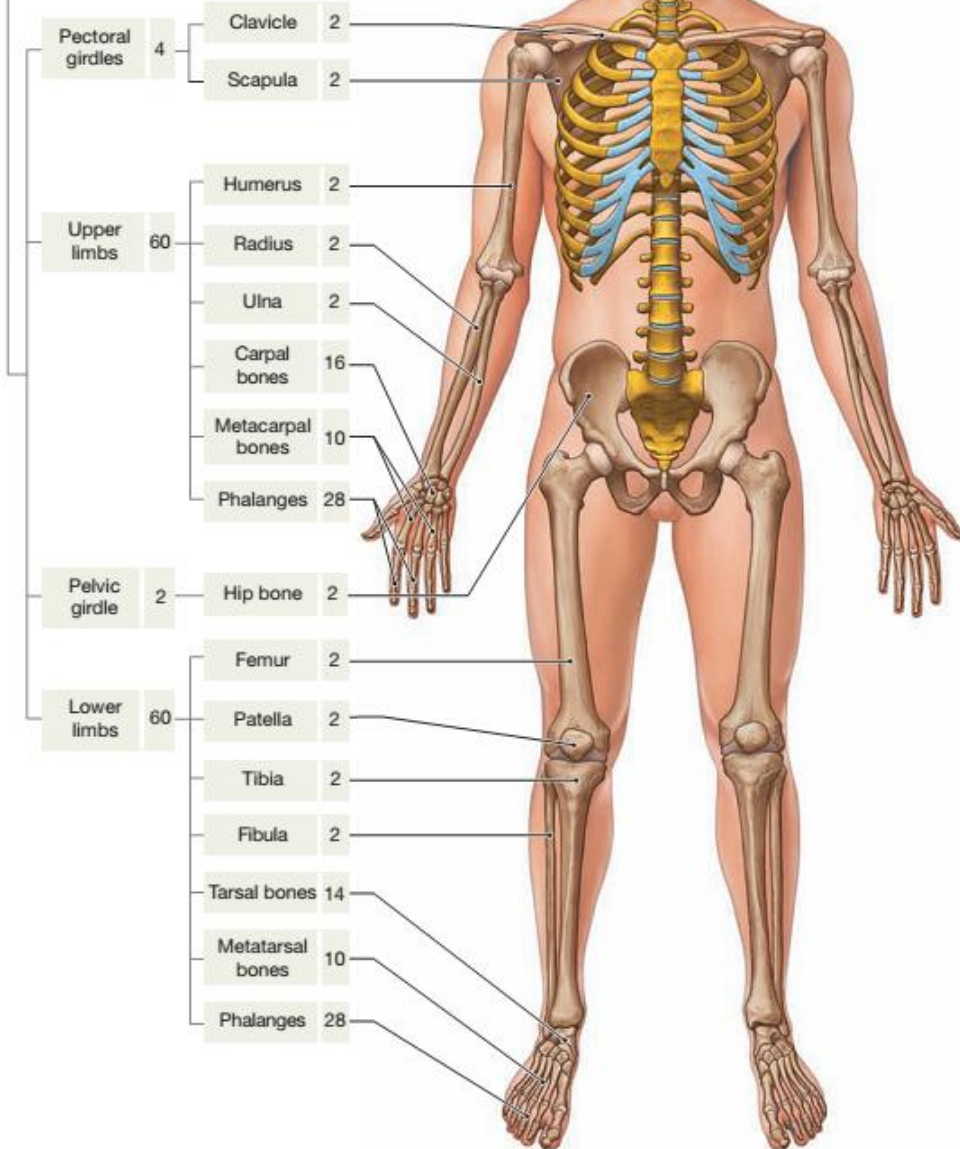
**c** Scoliosis



**a** An anterior view, showing the costal cartilages and the sternum



**b** A posterior view, showing the articulations of the ribs and vertebrae





□ **Thank you**